System Validation Report

Tester: Dasari Vinay
Build: branch `main` at merge commit `5b4873c` (pull request #34), plus the window program `hms_gui` and the fixes made during this run (the pull request that adds the window program), Visual Studio 2026 (MSVC 19.51), CMake 4.4.4, Debug configuration
Environment: Windows 11 Home, build 26300
Date: 10 November 2026

How it was run: `hms_gui.exe` (the window program, D-61) ran on a new, empty data store with made-up patients and staff. A script clicked and typed into the window, and every screenshot is a capture of that window only. The staff accounts and the doctors' weekly hours were created beforehand with the console program `hms.exe`, to save time; everything else, including the first start-up, the Administrator's account creation and the staff-account cases, was done in the window. TC-Perf-01 used the project's `hms_seed` tool. TC-Sec-01, TC-Sec-03 and TC-Sec-06 used small read-only checks on the data store file. The images for those checks and for the automated suite are the captured output of the command, drawn as text on a dark background; they are not photographs of a terminal window.

Only tests that were actually run are marked Pass, Partial or Fail. Tests not run are marked Not run, with the reason. Test case IDs are from the Test Plan (Section 13) and the SRS traceability table (Section 8). Where the build differs from the SRS, the row names the entry in `docs/deviation-log.md`.

Summary: 29 test cases, 28 run: all 28 passed, 1 not run (TC-UX-01). The Test Plan's own 18 cases: 17 passed, 1 not run. Four additional checks (X-01 to X-04) all passed. The automated suite passed: 281 test cases, 5,316 assertions.

Two defects in the window program were found by this run and fixed in the same change (see "Defects found in this run"). The console program was exercised only through the automated suite and the account set-up described above; the console screens were not walked through again.

Test cases

Test Plan cases

ID	Test	Expected	Actual	Result
TC-Auth-01	Weak and wrong passwords, then a valid login	Wrong password refused without saying which part was wrong; valid login opens the role's menu	First-run Administrator form refused a weak password; "Invalid username or password." for a wrong one; Receptionist login opened the Receptionist menu	Pass
TC-Sec-02	Log in as each of the five roles; a doctor opens a patient with no appointment	Each role gets only its own menu; out-of-role access refused and logged	Five separate menus shown; Dr rao refused the medical history and visit history of a patient he has no appointment with; the audit log holds `MEDICAL_HISTORY_VIEW` and `CONSULTATION_VIEW` entries with outcome DENIED (reason `NO_CARE_RELATIONSHIP`)	Pass
TC-Sec-03	Weak passwords at set-up; read the stored values	Weak passwords refused; salted hash stored, never plaintext	Refused (TC-Auth-01); all six stored values start with `$argon2id$`; two accounts given the same password have different values	Pass

ID	Test	Expected	Actual	Result
TC-Sec-04	Five wrong passwords, then the correct one, then Admin unlock	Locks on the fifth; correct password refused while locked; unlock works	Locked on the fifth ("Try again in 15 minutes"); correct password refused while locked; Admin saw status "locked" and unlocked; the login after the unlock worked	Pass
TC-Reg-01	Register a patient with a future birth date and an invalid mobile number, then correct both	Each invalid value refused; patient saved with a UHID	Both refused with a message naming the rule; details read back; HMS-000001 issued	Pass
TC-Reg-02	Register a person sharing a mobile number; register an existing patient again	Possible matches shown before saving; a different person gets a new UHID; the same patient gets no new record	SAME PHONE match shown; "different person" issued HMS-000002; a third patient got HMS-000003; registering Asha Kumar again listed STRONG (HMS-000001) and SAME PHONE (HMS-000002) matches	Pass
TC-Rec-02	Search by part of a name, shared mobile number, UHID, name and date of birth, and a name that does not exist	Matching patients found; no match gives a message	2, 2, 1 and 1 patients found; "No patient found."	Pass
TC-App-01	Book an appointment today and one tomorrow	Appointment saved with a number	APT-000001 (today 19:15) and APT-000002 (tomorrow 10:00) booked	Pass
TC-App-02	Book another patient with the same doctor; book the same patient at the same time with another doctor	A taken slot is not offered; a clash is refused	19:15 not in Dr mehta's free slots (20 free against 21 for Dr rao); "This patient already has an appointment at 10-11-2026 19:15 with Dr mehta (APT-000001). Nothing was saved."	Pass
TC-Doc-03	Record a consultation before the start, after the start, and a second time	Early and second attempts refused; valid one saved with medicines and tests	Early attempt refused on the first screen, before any form; CON-000001 saved with 1 medicine and 1 test and the appointment Completed; second attempt refused; CON-000002 saved for APT-000001 later	Pass
TC-Bill-01	Bill an unfinished visit; bill a completed visit with a test and a service charge	Unfinished visit refused; itemized bill with the correct total	Refused for APT-000001; BILL-000001 for Rs 850.00 (600 + 150 + 100)	Pass
TC-Bill-02	Part payment, overpayment, final payment	Balance updated; overpayment refused	Rs 300 cash leaves Rs 550; Rs 600 refused ("more than the outstanding balance"); Rs 550 card leaves Rs 0, bill Paid	Pass
TC-Bill-03	Receipt on payment and printed again	Receipt shows the payment and balance	RCPT-000001 and RCPT-000002 issued; RCPT-000001 shown again with the same amounts. The window program shows the receipt on screen only; the text-file save is console-only (D-46, D-61)	Pass
TC-Lab-01	Enter a result, correct it, view as the ordering doctor and as another doctor	Result saved and visible to the treating doctor	Revisions 1 and 2 saved; ordering doctor sees revision 2; other doctor sees nothing	Pass

ID	Test	Expected	Actual	Result
TC-Pharm-01	Add medicines, dispense, dispense more than prescribed, restock	Stock reduces by the quantity; over-dispensing refused; low-stock alert below the threshold	Three medicines added; LOW STOCK shown at 4 and 8 (threshold 10, D-54); picking Paracetamol against an Amoxicillin line showed a warning and kept Dispense disabled until confirmed (D-53); 4 Amoxicillin dispensed, line complete, stock 0 with a LOW alert; 11 of 10 refused; 10 Paracetamol dispensed, stock 100 to 90; restock took Amoxicillin from 0 to 20 and cleared the alert	Pass
TC-Perf-01	Search and book with 100,000 patients, 100 runs per operation (Debug build, encrypted store)	Search and booking screens: 90th percentile at most 3 s	90th percentile: UHID 0.24 ms, name 84.7 ms, phone 0.18 ms, booking screen 1.45 ms, find-and-book a slot 9.05 ms. Timings exclude display and typing. A Release build was not timed	Pass
TC-Sec-01	After the full run, check the audit entries and the store file	No plaintext PII or medical data stored	The store file does not start with `SQLite format 3`; entropy 7.9994 bits per byte; none of 24 patient, clinical and account strings found in it. No audit entry mentions patient text (0 matches); entries hold IDs and field names	Pass
TC-UX-01	3 new receptionists each register a patient after 30 minutes of training	Average at most 5 minutes	Not run: to be done by three members of the team (D-32)	Not run

SRS traceability cases

ID	Test	Expected	Actual	Result
TC-Rec-01	Doctor records a medical history, updates one field, views earlier versions	Update timestamped; prior version kept	MHV-000001 and MHV-000002 saved with time and doctor; only the allergies changed; MHV-000001 listed unchanged	Pass
TC-App-03	Reschedule and cancel outside the cutoff; cancel inside the cutoff (D-38)	Changes work outside the cutoff; slot released; refused inside it	APT-000002 moved to APT-000003 and its slot freed; APT-000003 cancelled; APT-000001 refused ("changes close 60 minutes before the start")	Pass
TC-App-04	Open the reminder list twice	A reminder entry for every booked appointment in the window (48 hours, D-40)	Both appointments listed NEW the first time and "listed before" the second	Pass
TC-Doc-01	Admin sets a doctor's department and weekly hours	Doctor appears on the booking screen	Dr mehta's profile and weekly hours shown; both doctors offered on the booking screen on every working day	Pass
TC-Doc-02	Doctor opens today's queue	Sorted by time; only that doctor's patients	APT-000001 shown with patient and status	Pass
TC-Staff-01	Deactivate an account, try to log in, deactivate own account, reactivate, set a new password	Deactivated account cannot log in; action audited	Login refused; own account refused; reactivation and a new password worked, and the lab technician logged in with it; the actions are in the audit log	Pass

ID	Test	Expected	Actual	Result
TC-Admin-01	Add a department and a room; view the doctor-department mapping	Changes reflected immediately in booking and billing	Department and room saved and listed; the two doctors' departments (set with their weekly hours, not typed into the department list) appear in the department list with one doctor each (D-59); booking shows "Dr mehta, General Medicine" and the bill shows "Consultation - Dr rao (Cardiology)"	Pass
TC-Admin-02	View and update system parameters	Value changes and is used without a rebuild	The three parameters shown; cutoff changed from 60 to 45 and audited with the old and new value; the change-appointment screen then read "Changes close 45 minutes before"; a cancel 59 minutes before the start, which the earlier 60-minute rule refused (TC-App-03), was accepted (D-60)	Pass
TC-Rep-01	Summary report for 04-11-2026 to 11-11-2026	Totals match the data entered	3 patients registered; 4 appointments (0 booked, 2 completed, 2 cancelled); 2 consultations; 1 bill of Rs 850.00; payments Rs 850.00 (cash Rs 300.00, card Rs 550.00); Rs 0.00 unpaid. No No-show column (D-35)	Pass
TC-Sec-05	View the newest entries and the refusals in the audit log	Every action produces a matching entry	Every action of the run is listed; denied logins, the lockout and the unlock appear with reason codes; the settings entry shows the old and new value	Pass
TC-Sec-06	Read the Windows permissions of the store	Access limited to the application's own user; sensitive fields encrypted	A store created by the window program: folder, store file and lock file each list only the running user; the file is encrypted (TC-Sec-01) (D-06, D-07)	Pass

Additional checks

These cover behaviour defined in the deviation log or added after the Test Plan, which has no test case ID in the Test Plan or SRS.

ID	Test	Expected	Actual	Result
X-01	Visit history (HMS-F-006, D-49): treating doctor and a doctor with no appointment	Treating doctor sees the visits; other doctor refused	CON-000001 shown in full to Dr rao; Dr rao refused for the patient he has no appointment with	Pass
X-02	Start-up (HMS-F-053, D-21): no store, decline; damaged store	Nothing created without confirmation; damaged file refused and unchanged	Quit on the new-store question created no folder; a damaged file was refused after three passphrase attempts with the same SHA-256 before and after	Pass
X-03	Single instance (SRS 2.4): second instance on the same store, then close the first	Second instance refused; next start works after the first ends	"Cannot start: the data store is already in use by another instance of the application. Nothing was changed."; the next start reached the passphrase screen	Pass
X-04	Correct a patient's details (HMS-F-005)	Changed field saved after read-back; audit names the field, not the value	Address of HMS-000003 changed after a read-back; the audit entry `PATIENT_UPDATE_DEMOGRAPHICS` lists `address` only	Pass

Not run, with reason: TC-UX-01 has not been done yet; three members of the team will be the testers (D-32). The Test Plan's exit criterion is the one adopted in D-32.

Evidence

The tables above give the result of every step. The images show the key steps of each case, not every step.

Automated suite

Not a Test Plan case; run on the same build for reference. Result: 281 test cases and 5,316 assertions passed, 0 failed. Output of the command, drawn as an image.

```
> ctest --test-dir build -C Debug --output-on-failure
Test project F:/Sem 5/SE/Mini Project/Hospital-Management-System/build-gui
    Start 1: hms_tests
1/1 Test #1: hms_tests .......................   Passed   80.63 sec

100% tests passed out of 1

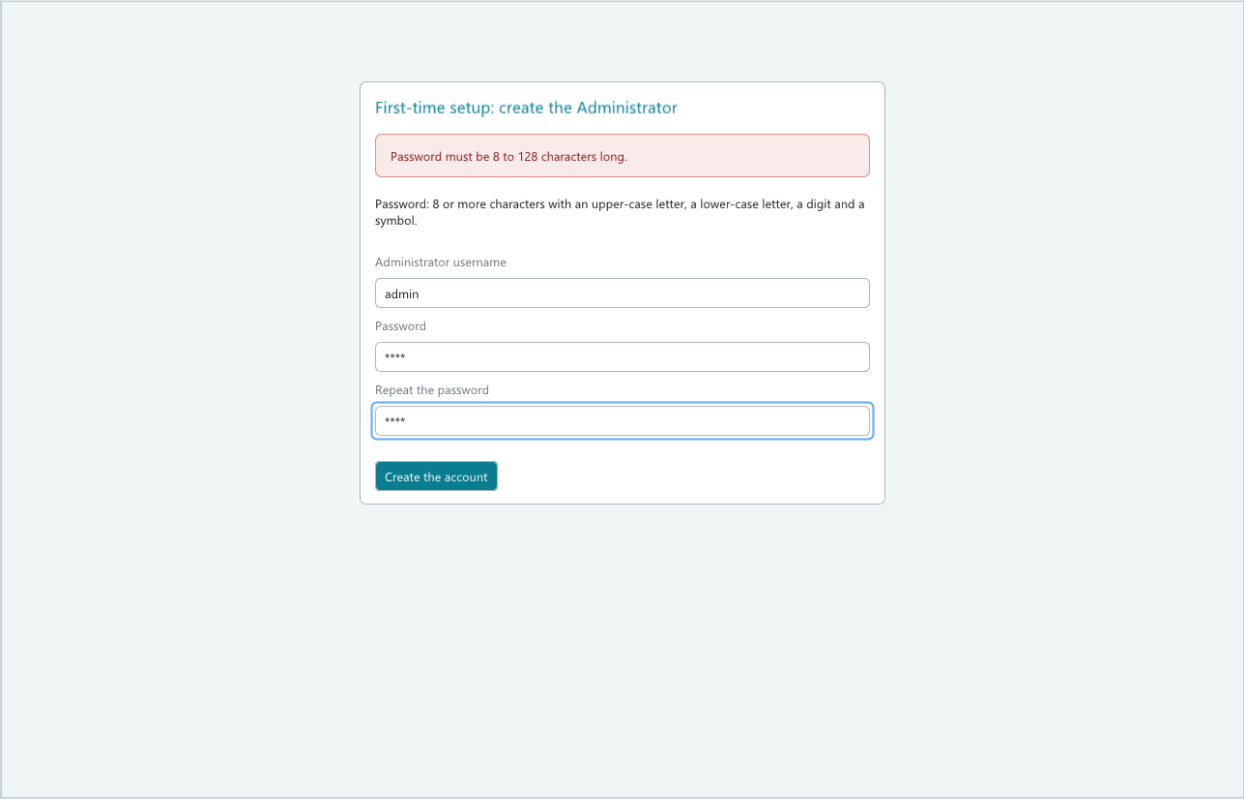
Total Test time (real) =  80.65 sec

> build\Debug\hms_tests.exe  (same tests, doctest summary)
[doctest] doctest version is "2.4.11"
[doctest] run with "--help" for options
===============================================================================
[doctest] test cases:  281 |  281 passed | 0 failed | 0 skipped
[doctest] assertions: 5316 | 5316 passed | 0 failed |
[doctest] Status: SUCCESS!
```

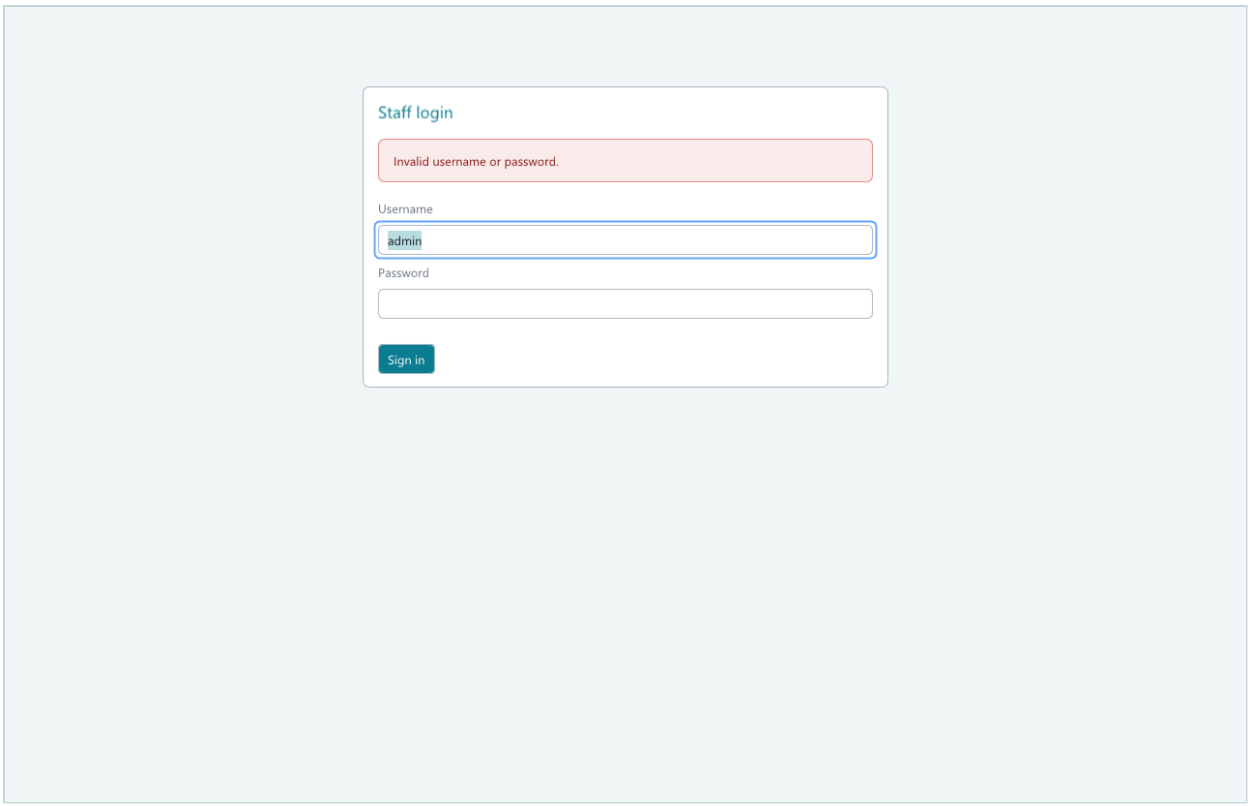
Automated suite passed

TC-Auth-01: Login

A weak password refused at the first-run Administrator form and a wrong password at login. The menu after a valid login is in the image under TC-Sec-02.



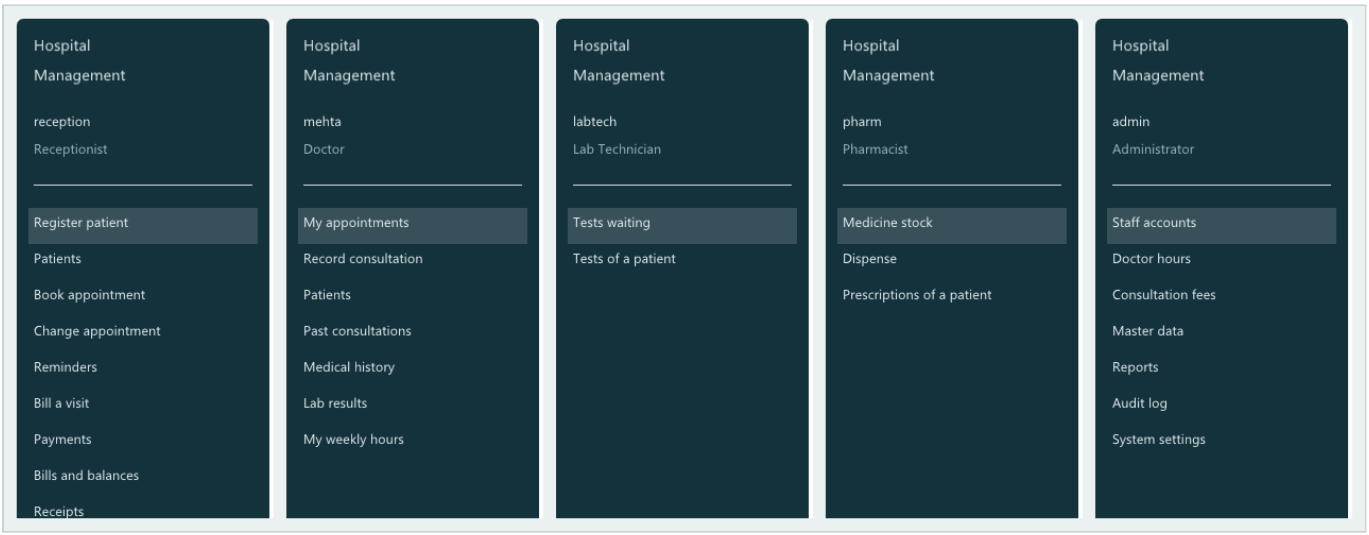
TC-Auth-01 weak password refused



TC-Auth-01 wrong password

TC-Sec-02: Role-based access

Each role gets only its own menu. The image shows the side menu of each of the five roles after login, from left to right: Receptionist, Doctor, Lab Technician, Pharmacist, Administrator. The refused doctor is under X-01 and TC-Sec-05.



TC-Sec-02 menus of the five roles

TC-Sec-03: Password policy and hashing

Weak passwords are refused in the TC-Auth-01 screenshot. The stored values are salted Argon2id hashes, and two accounts with the same password have different values. Output of a read-only check, drawn as an image.

```
> read each staff account's stored password value from the data store

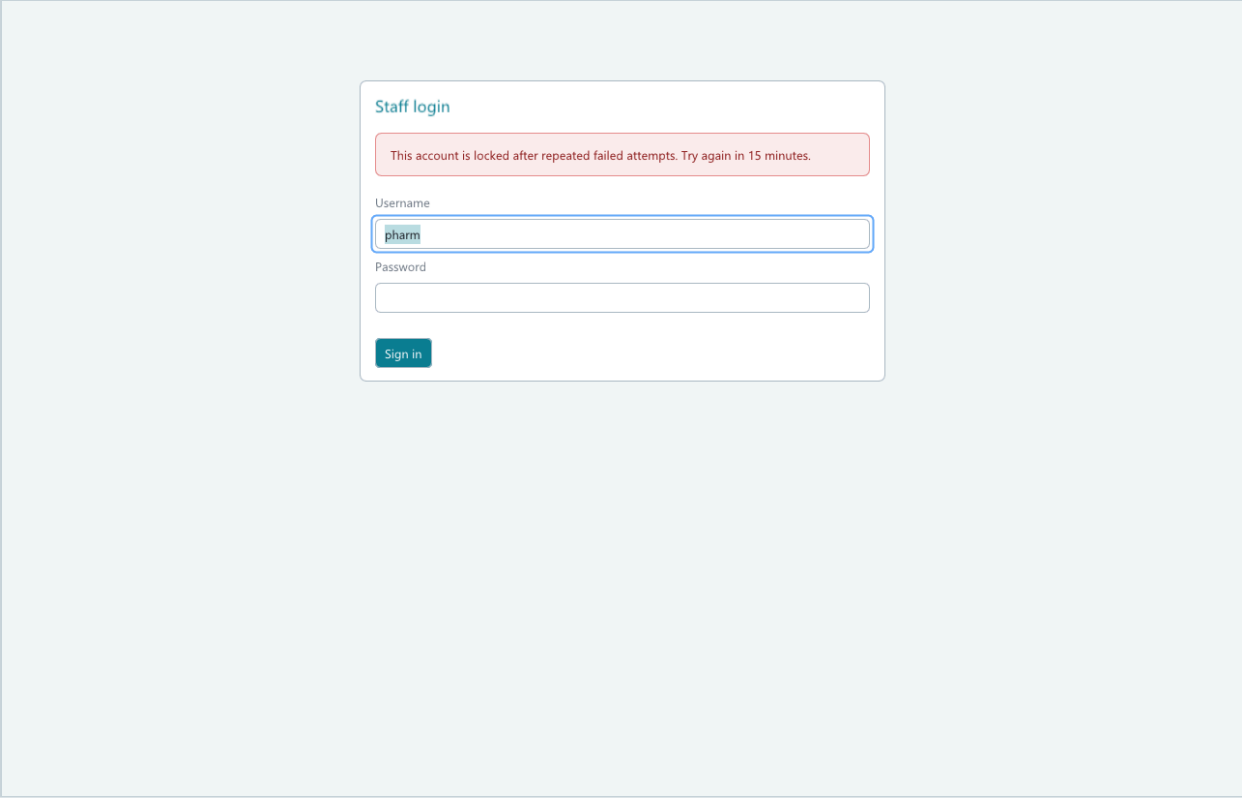
Account    Role            Stored value
admin      administrator   $argon2id$v=19$m=19456,t=2,p=1$TNCXJ66UW98oI7DUTnvhpQ$+OG8gXoxbU2g0iBxALfCrD5BHTOsj1JrPUAYfcGx0bY
labtech    lab_technician  $argon2id$v=19$m=19456,t=2,p=1$xLre/ddp0oW+NIwEvXmvwQ$2yR06SBR/0Ywypfnv7jjoREE/m84LCfE/FNIiN6lnz4
mehta      doctor          $argon2id$v=19$m=19456,t=2,p=1$Ww2jdZ2vjMK66+3xcTFOWw$uaIwuAsEU18rBxHkJBO30XrPr0ZY/RG5MmxQIx4L8K0
pharm      pharmacist      $argon2id$v=19$m=19456,t=2,p=1$n1WURq22QgBPmLo/yPLsPw$Tf/5W4s4RCbOpqFMjNk81xyJ3N/QE2j/COWpU35UPmo
rao        doctor          $argon2id$v=19$m=19456,t=2,p=1$J/pcvvVwt7mkFwFmZdWM/Q$IzPZ0ToHEhsDnAJ6MNOtB7YUsJw+t59cuG/8plSSqJg
reception  receptionist    $argon2id$v=19$m=19456,t=2,p=1$gL7Ss4qPwGD6UicVd7HjKQ$s/shEpqPfPrj1+c2Ym+l9HgtcFk6D+9OIEkz4VCIl10

mehta and rao were given the same password. Their stored values differ (random salt per account).
Every value starts with $argon2id$; no plaintext password is stored.
```

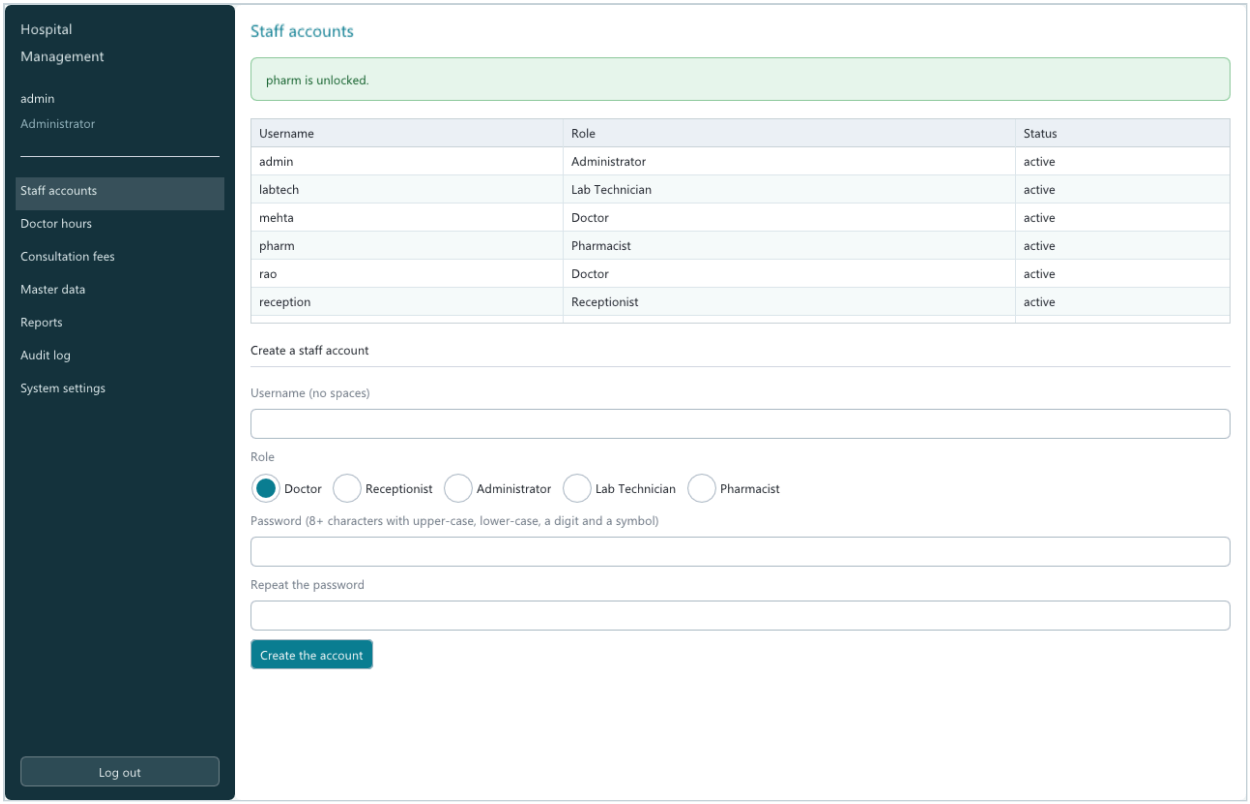
TC-Sec-03 salted Argon2 hashes

TC-Sec-04: Account lockout

Locked on the fifth wrong password, the correct password refused while locked, then unlocked by the Admin. The account also unlocks by itself after the 15-minute cooldown (D-16); this run used the Admin unlock.



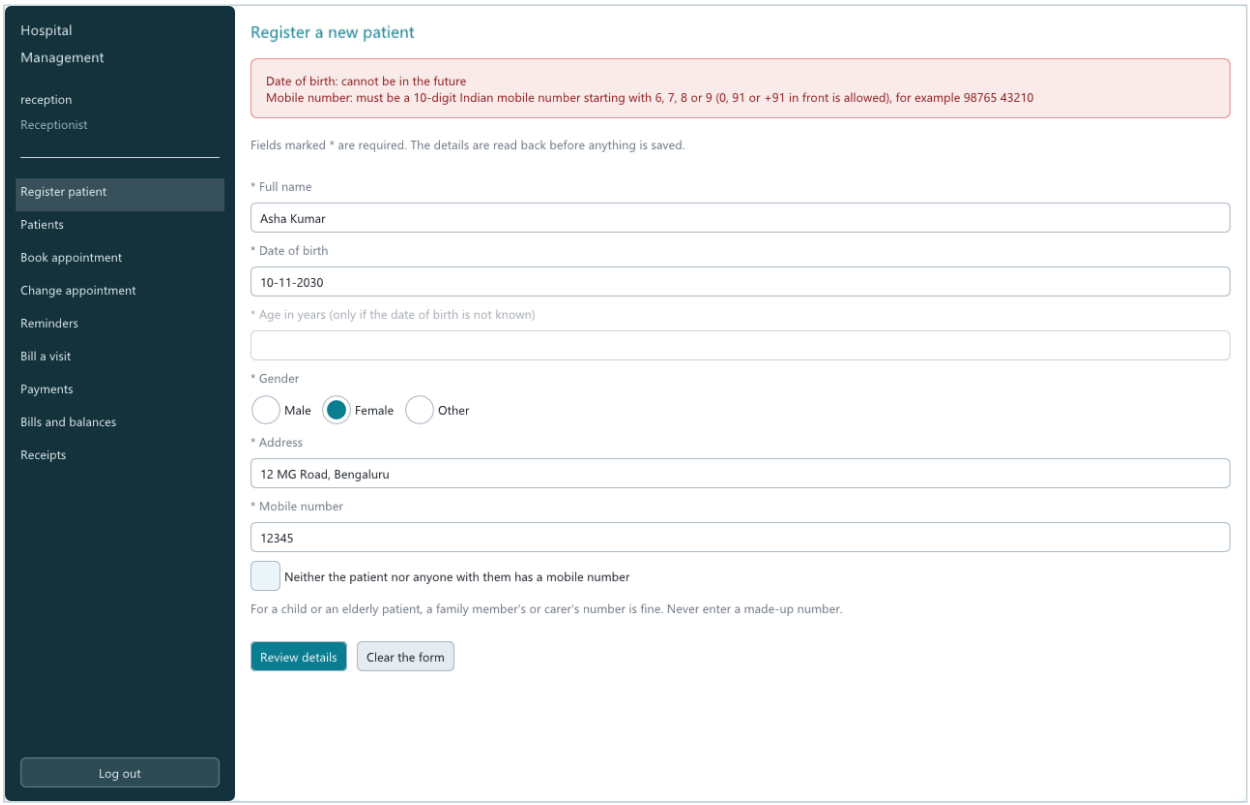
TC-Sec-04 correct password refused while locked



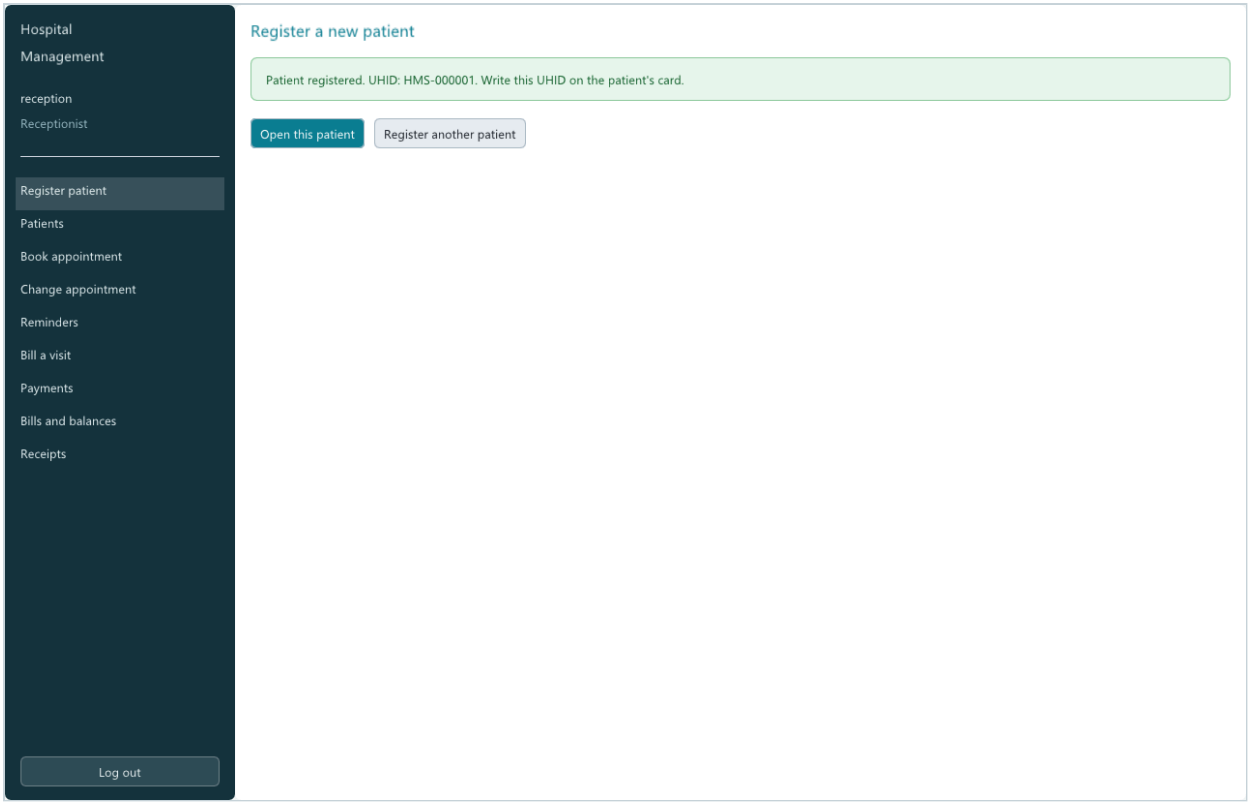
TC-Sec-04 Admin unlock

TC-Reg-01: Register a patient

A future birth date and a 5-digit mobile number are refused, then the corrected details are read back and the patient is saved.



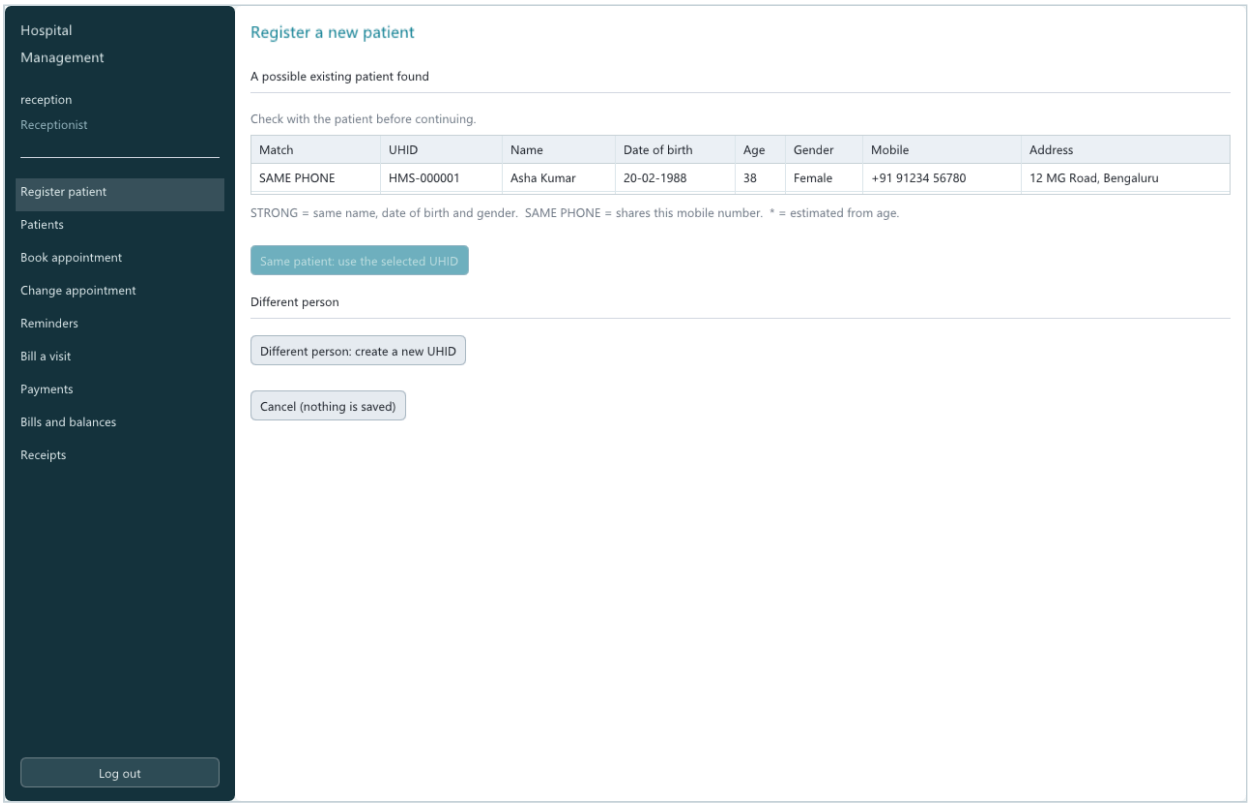
TC-Reg-01 validation errors



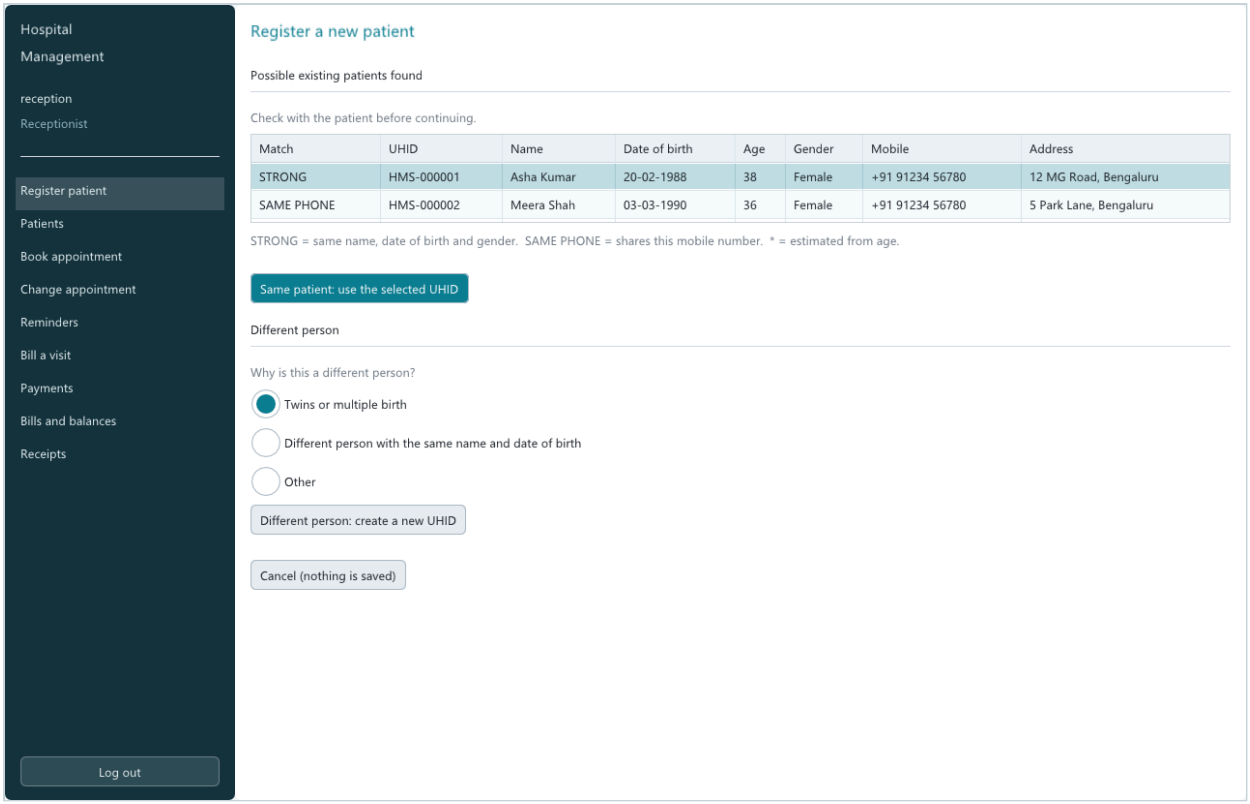
TC-Reg-01 saved

TC-Reg-02: Unique UHID and duplicate-phone detection

A shared mobile number shows a SAME PHONE match, and "different person" issues a new UHID. Registering an existing patient again shows a STRONG match together with the SAME PHONE match.



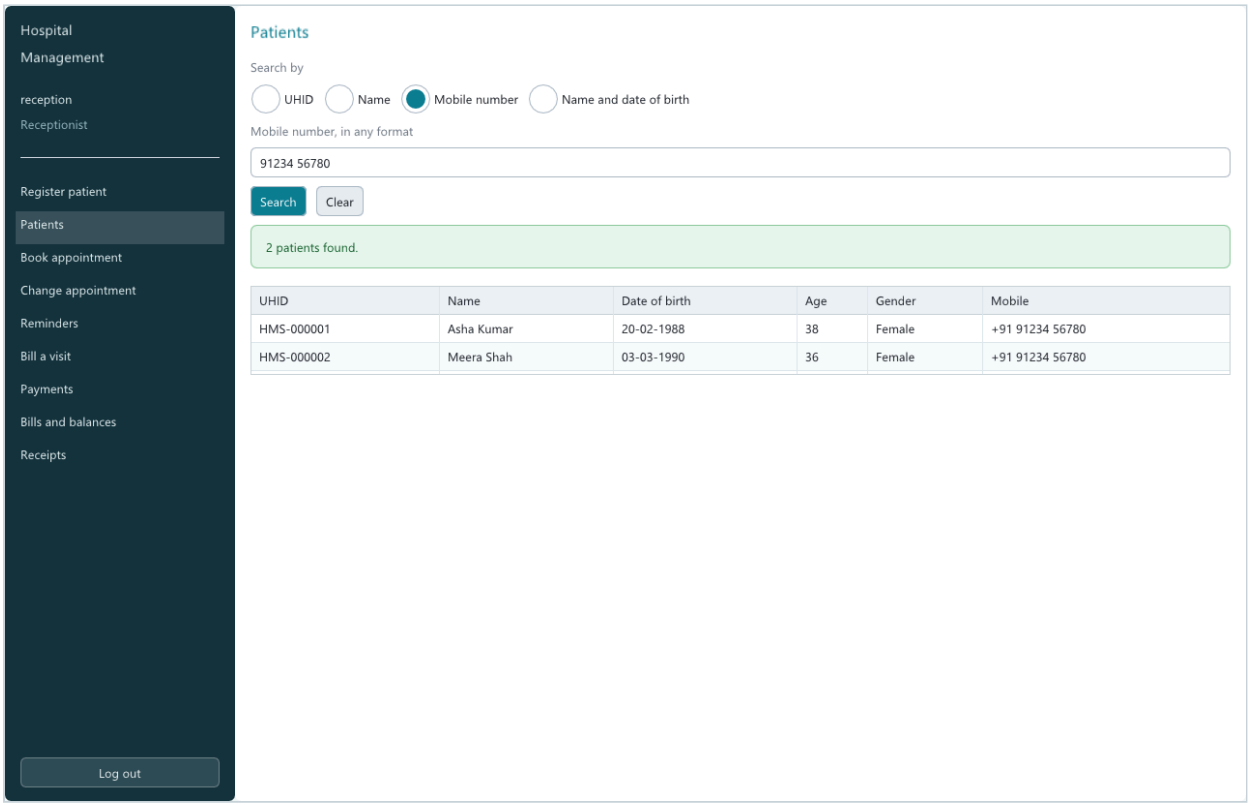
TC-Reg-02 shared phone detected



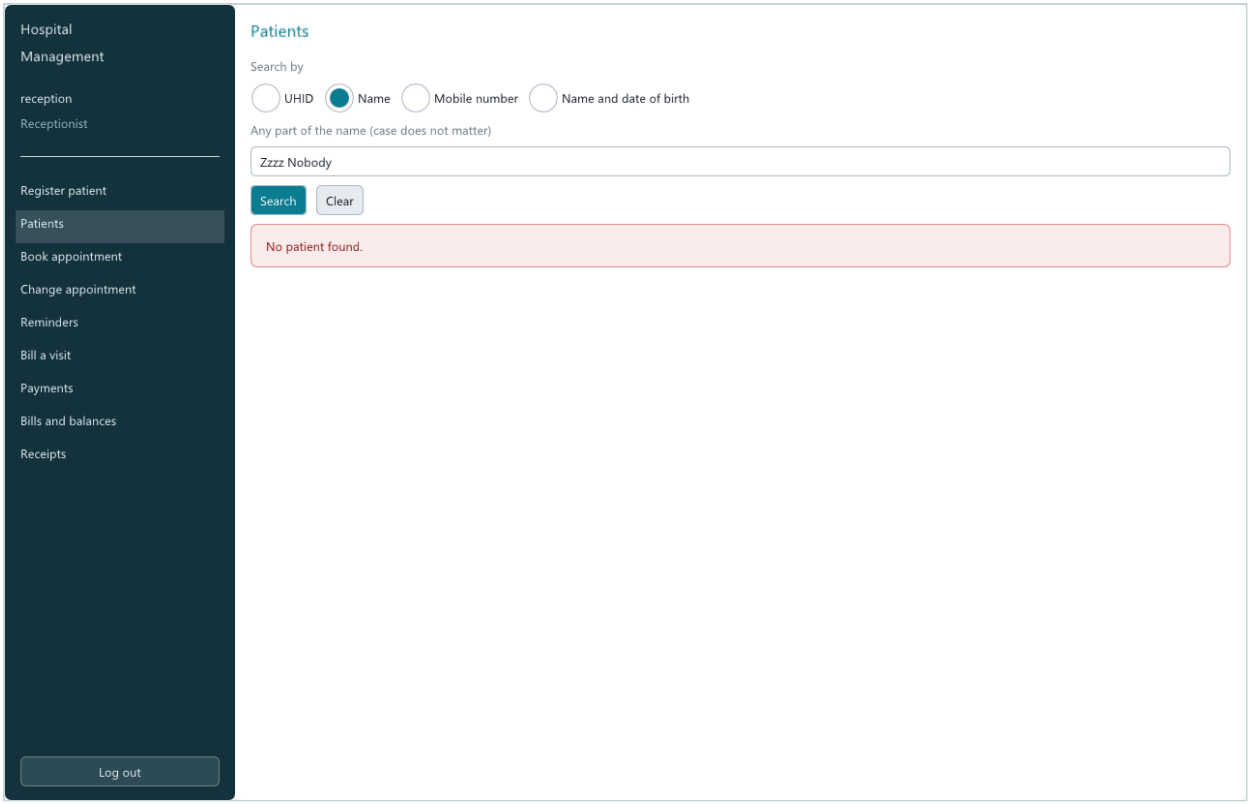
TC-Reg-02 strong match

TC-Rec-02: Search

By part of a name, shared mobile number, UHID, name and date of birth, and a name with no match.



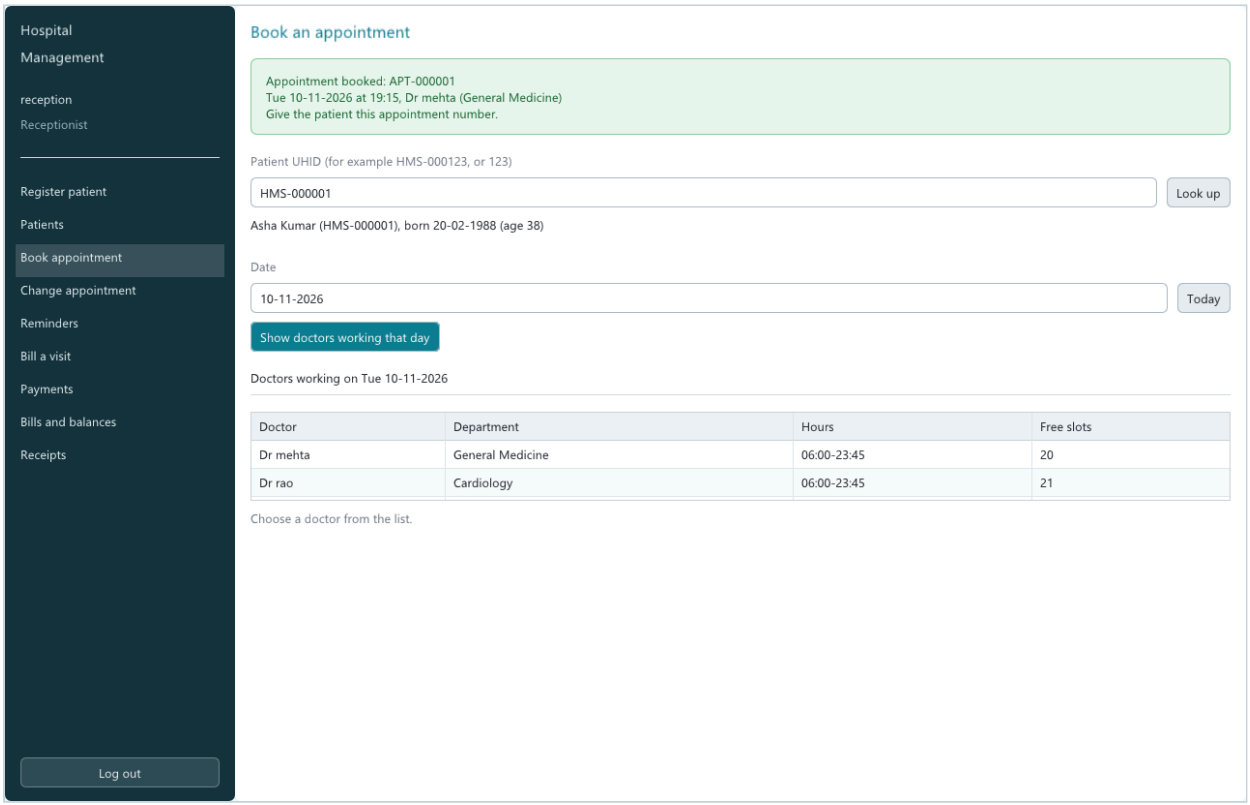
TC-Rec-02 shared mobile



TC-Rec-02 no match

TC-App-01: Book an appointment

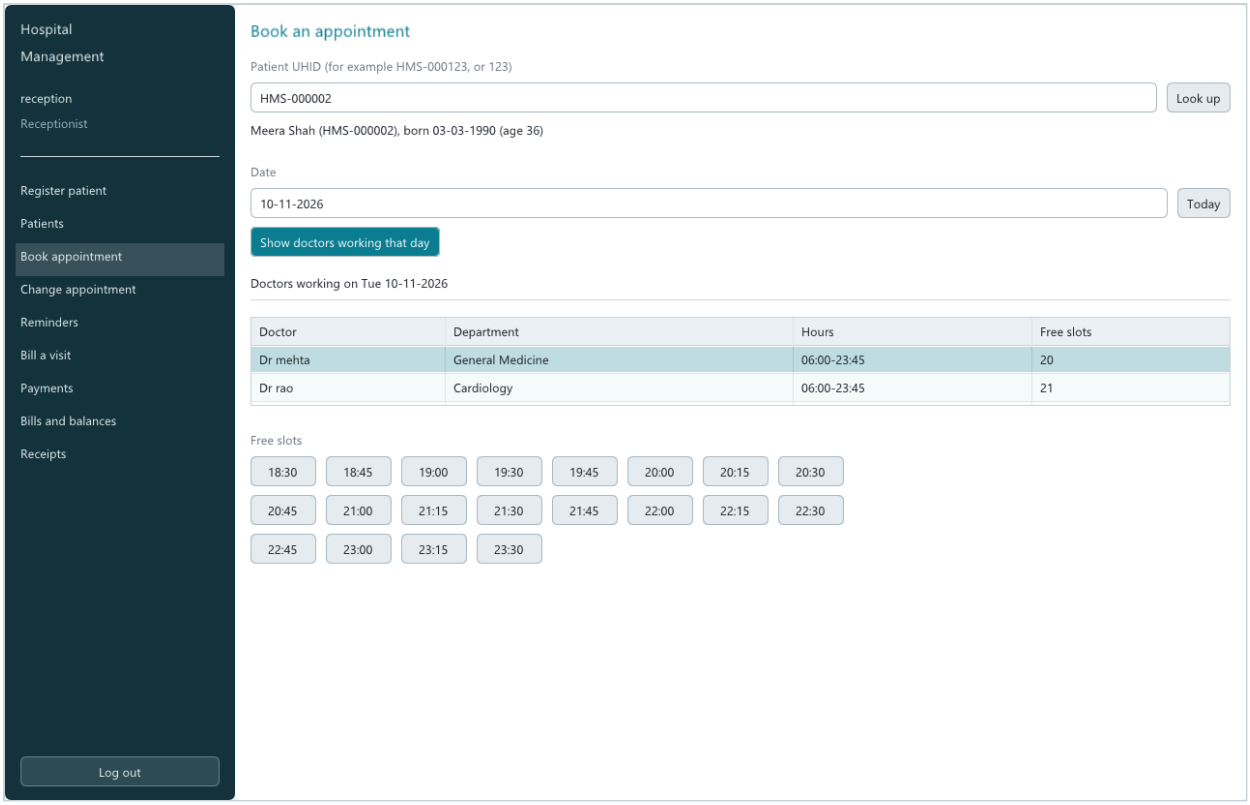
Booked for today and for tomorrow. The slots offered start at the next quarter hour, so the first free slot of the day depends on the time of the run.



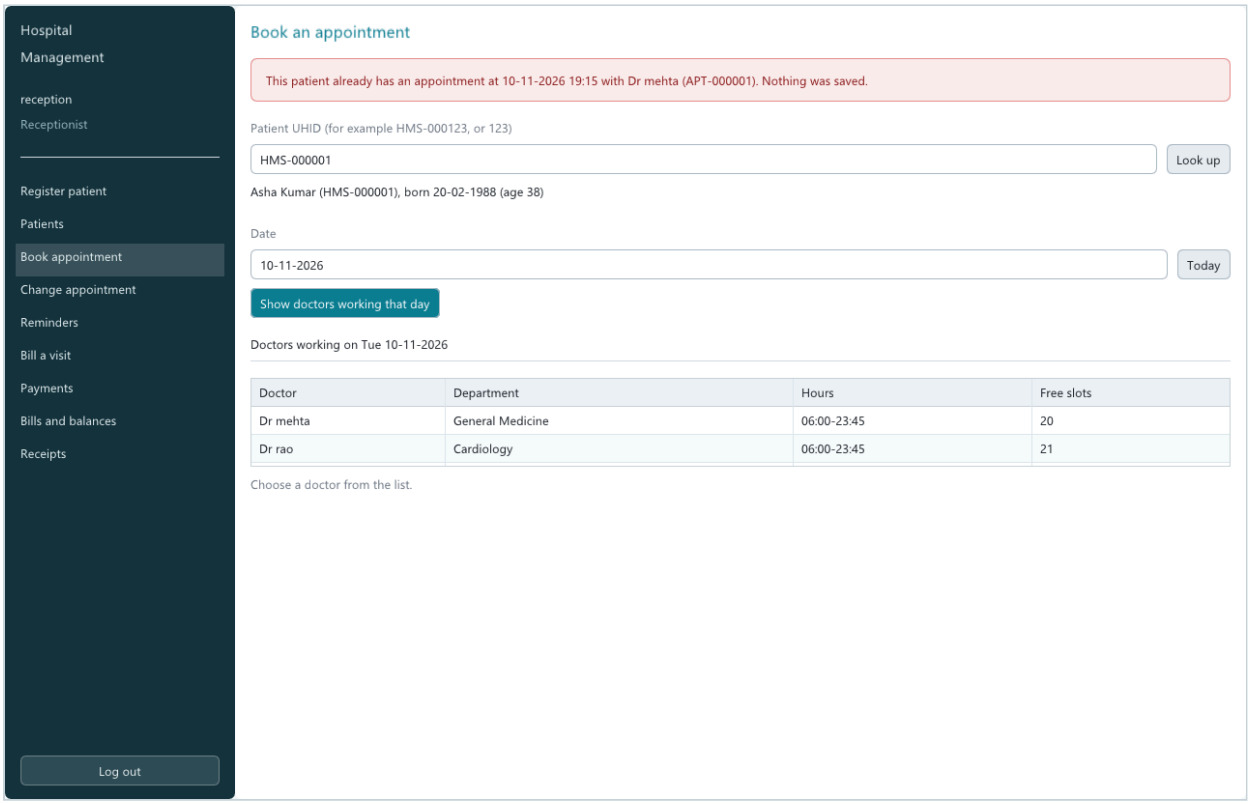
TC-App-01 book today

TC-App-02: Prevent double-booking

The taken slot is not offered to the next patient, and the same patient at the same time with another doctor is refused.



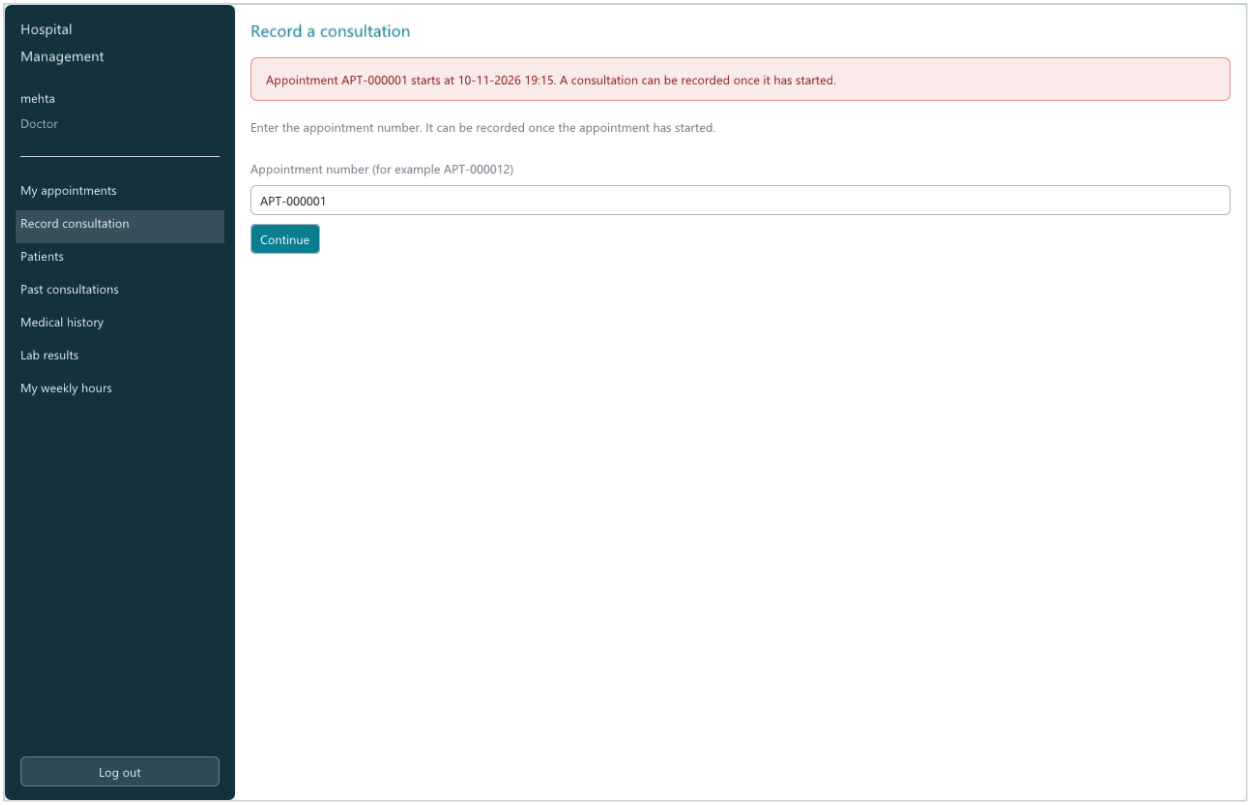
TC-App-02 taken slot not offered



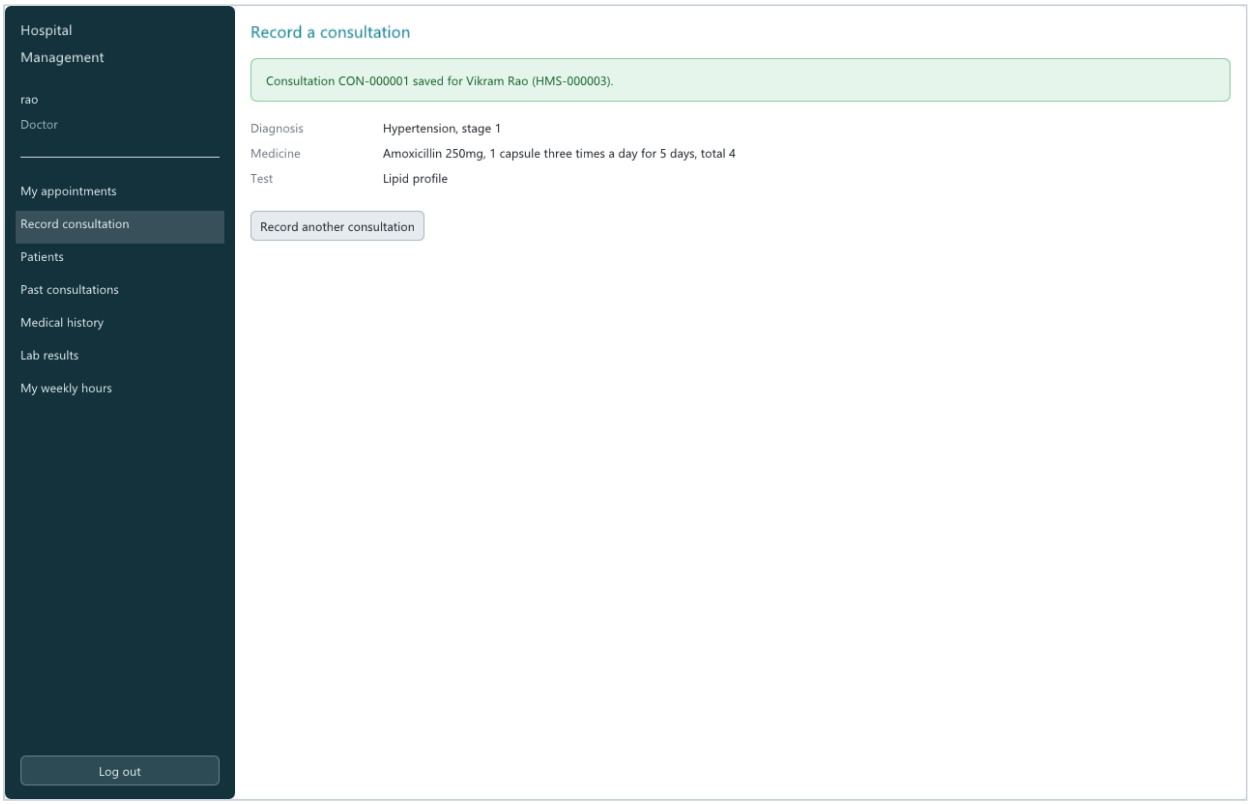
TC-App-02 patient clash refused

TC-Doc-03: Record a consultation

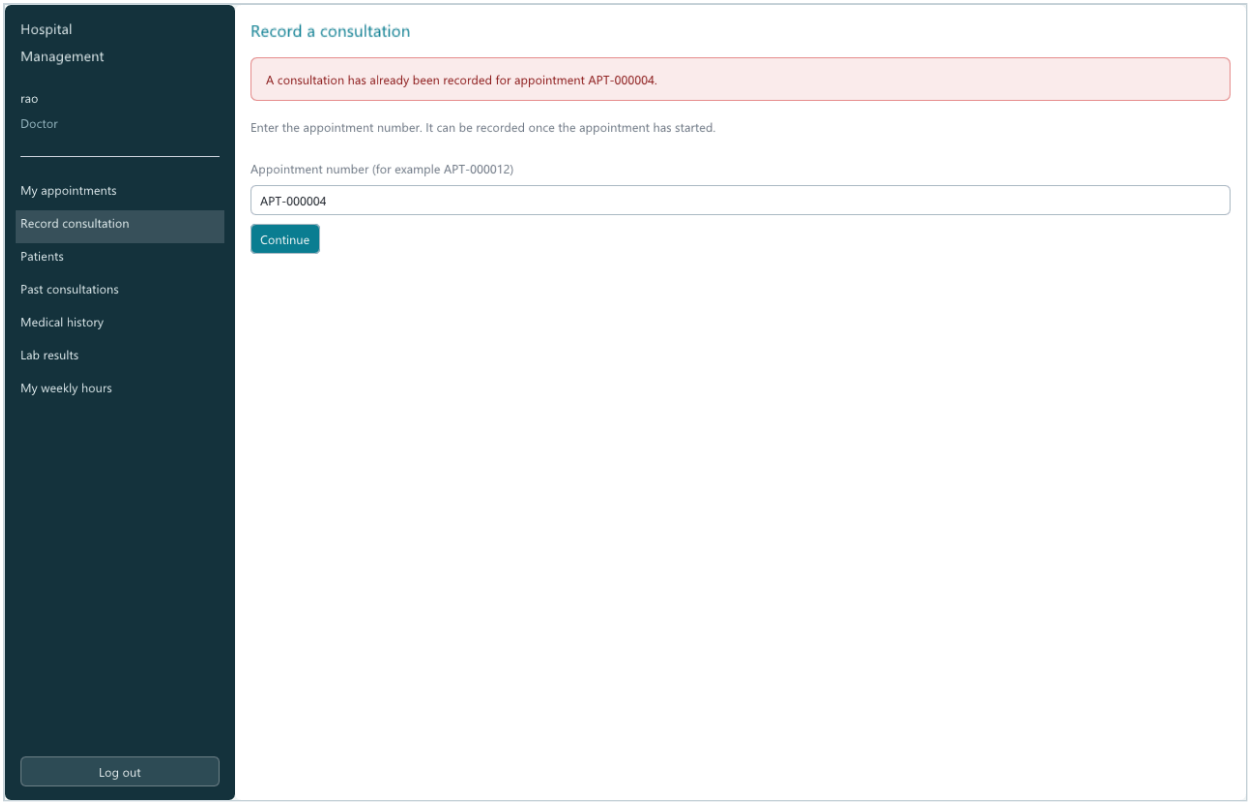
Refused before the start (on the first screen, before any form), saved after the start with a medicine and a test, and refused a second time. A second consultation (CON-000002) was saved for APT-000001 later in the run.



TC-Doc-03 before start refused



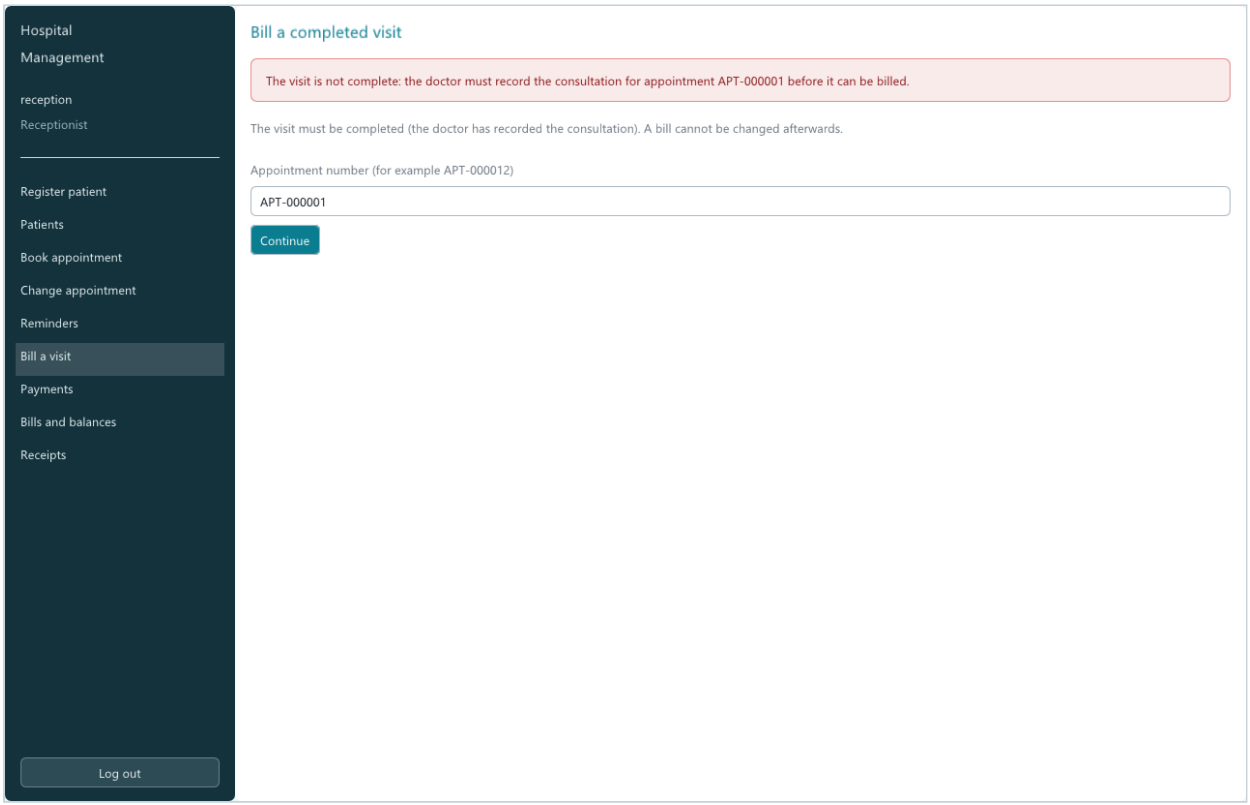
TC-Doc-03 consultation recorded



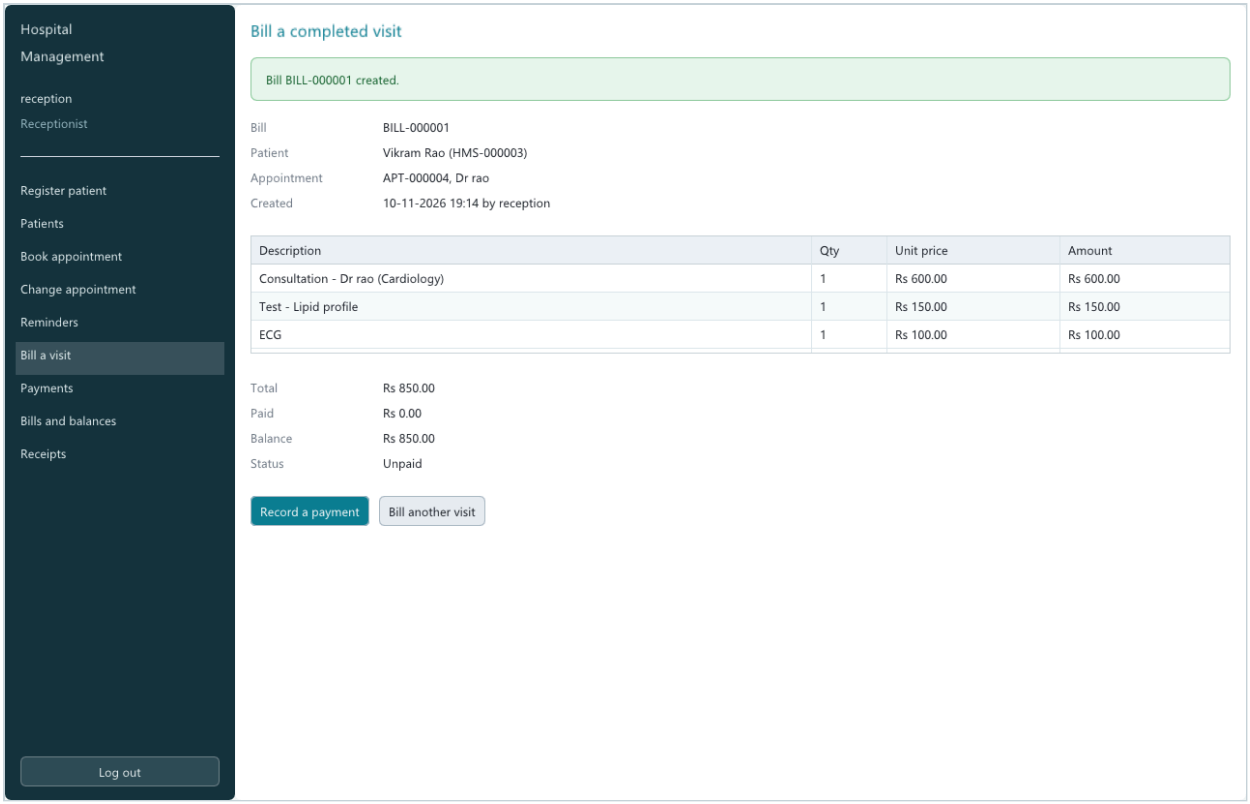
TC-Doc-03 second consultation refused

TC-Bill-01: Itemized bill

An unfinished visit is refused. A completed visit is billed with the consultation fee, a test price and a service charge.



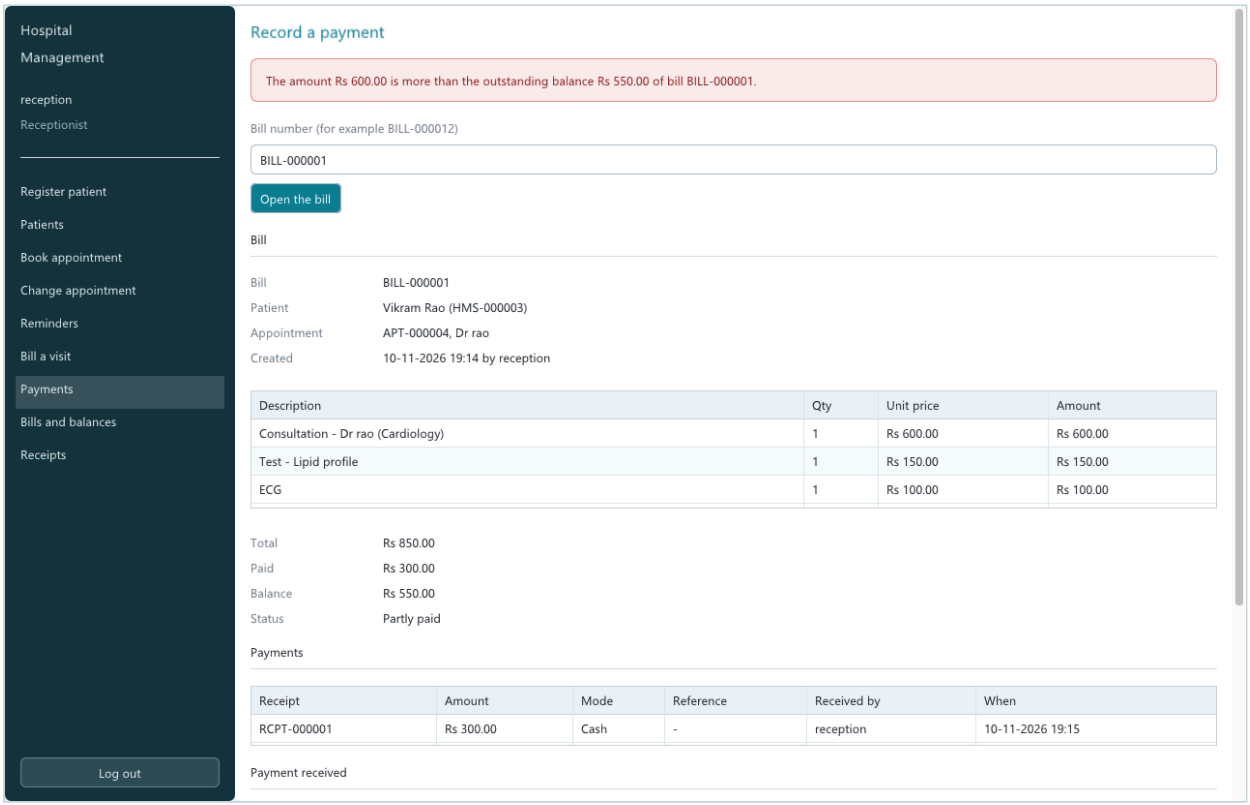
TC-Bill-01 unfinished visit refused



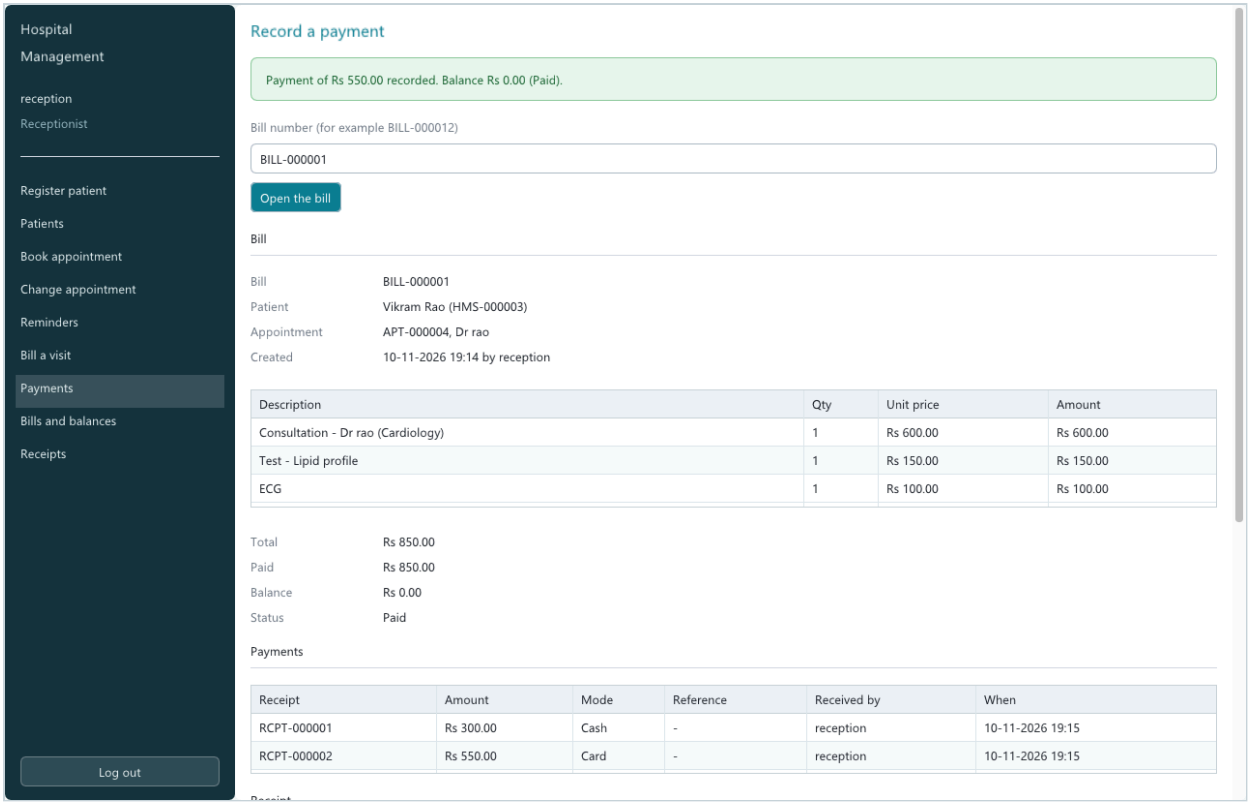
TC-Bill-01 itemized bill

TC-Bill-02: Payment

A part payment, an overpayment refused, and the final payment.



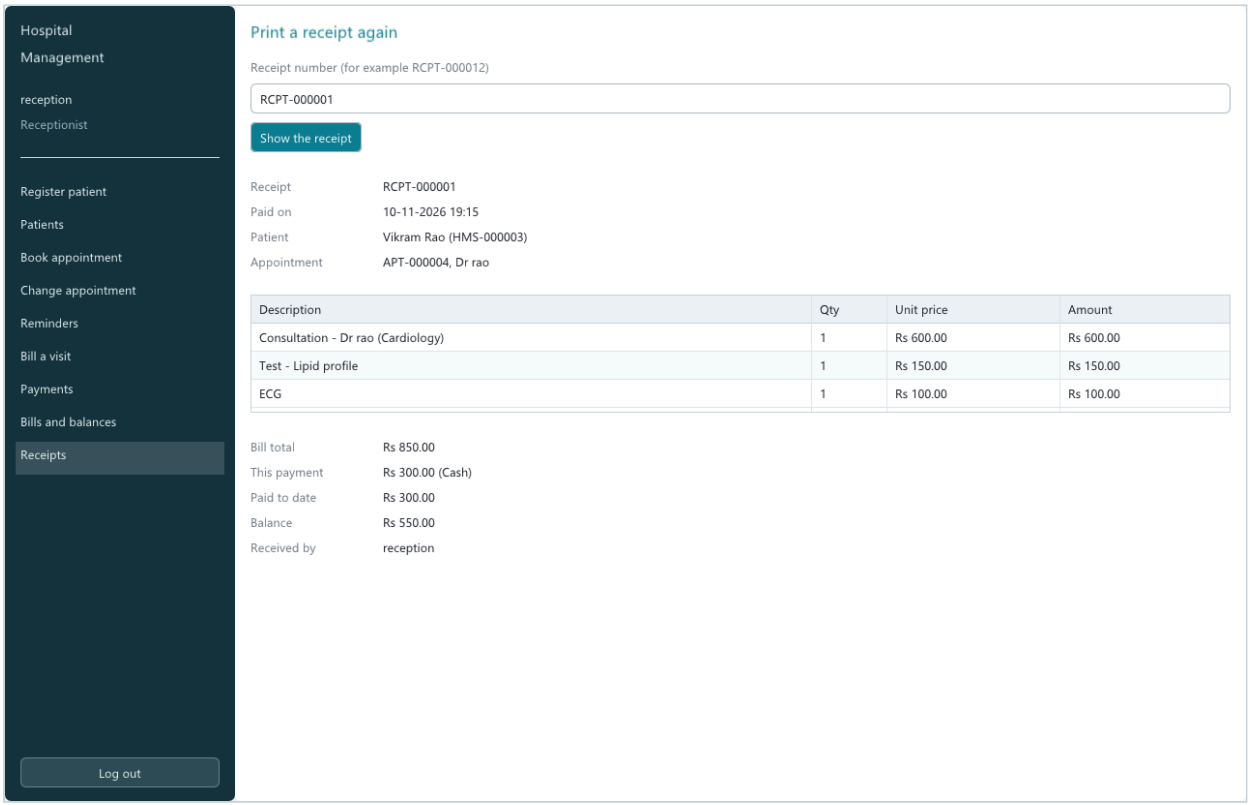
TC-Bill-02 overpayment refused



TC-Bill-02 final payment

TC-Bill-03: Receipt

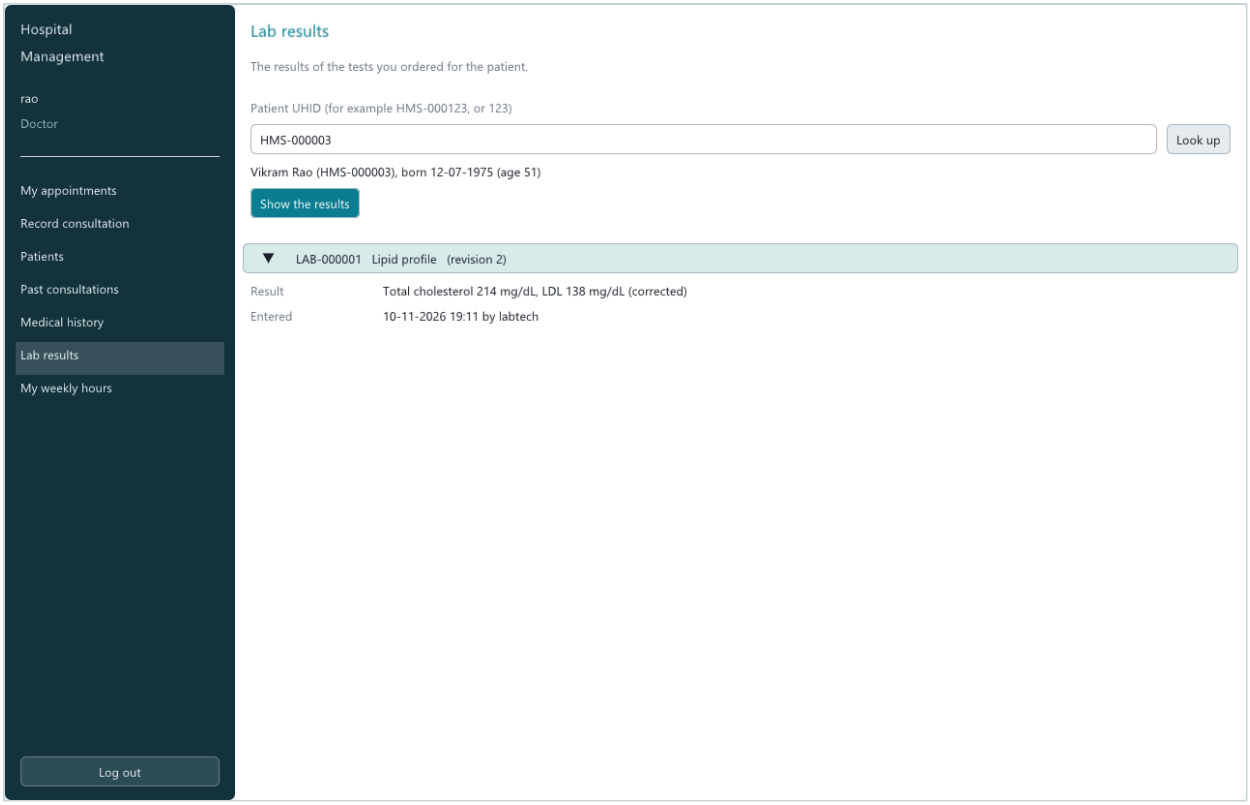
A receipt printed again.



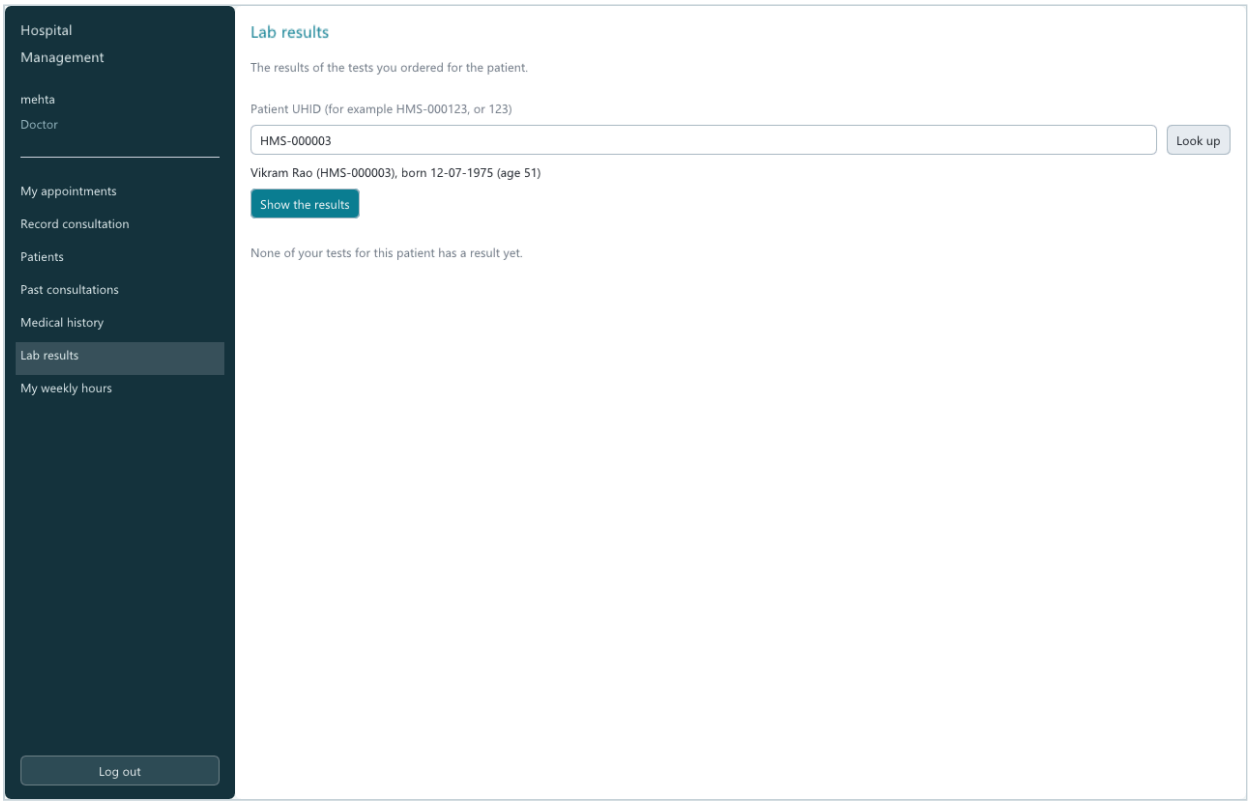
TC-Bill-03 receipt printed again

TC-Lab-01: Lab results

The waiting test, the first result, the corrected result, then the result as seen by the ordering doctor and by another doctor.



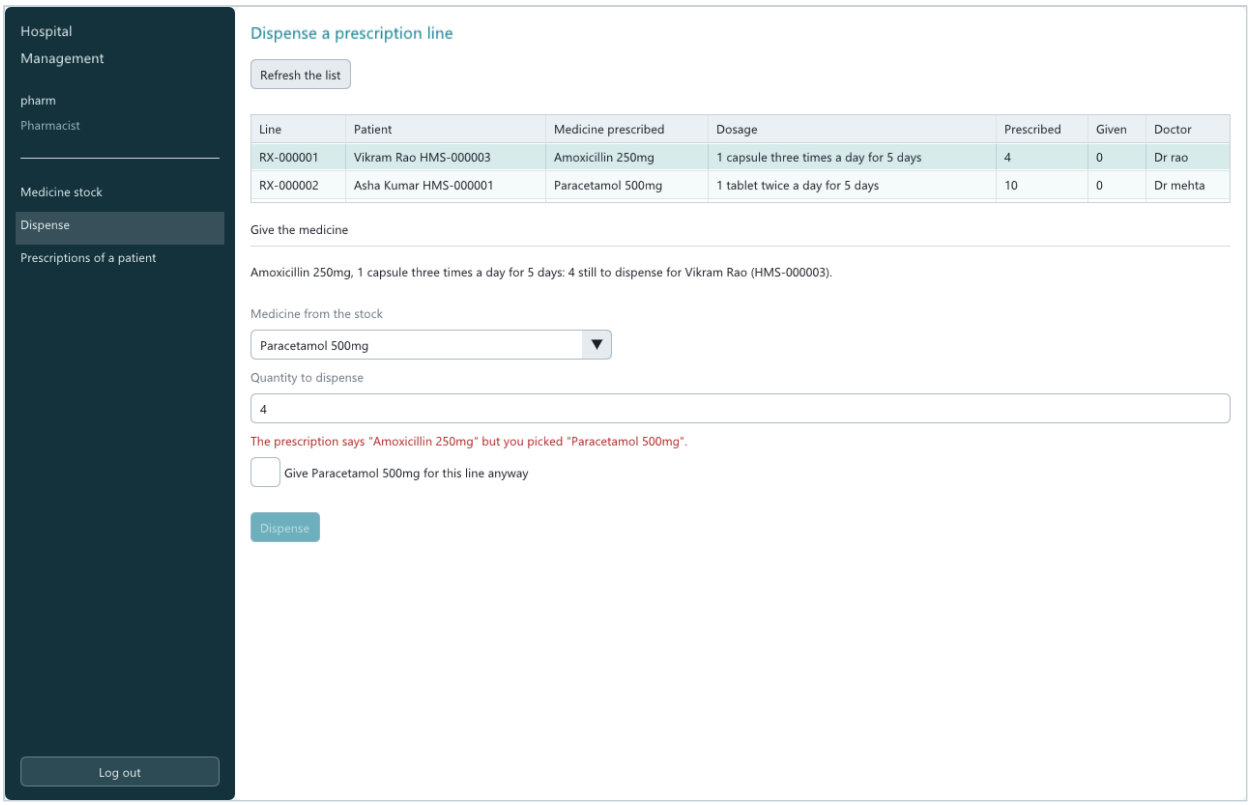
TC-Lab-01 ordering doctor sees result



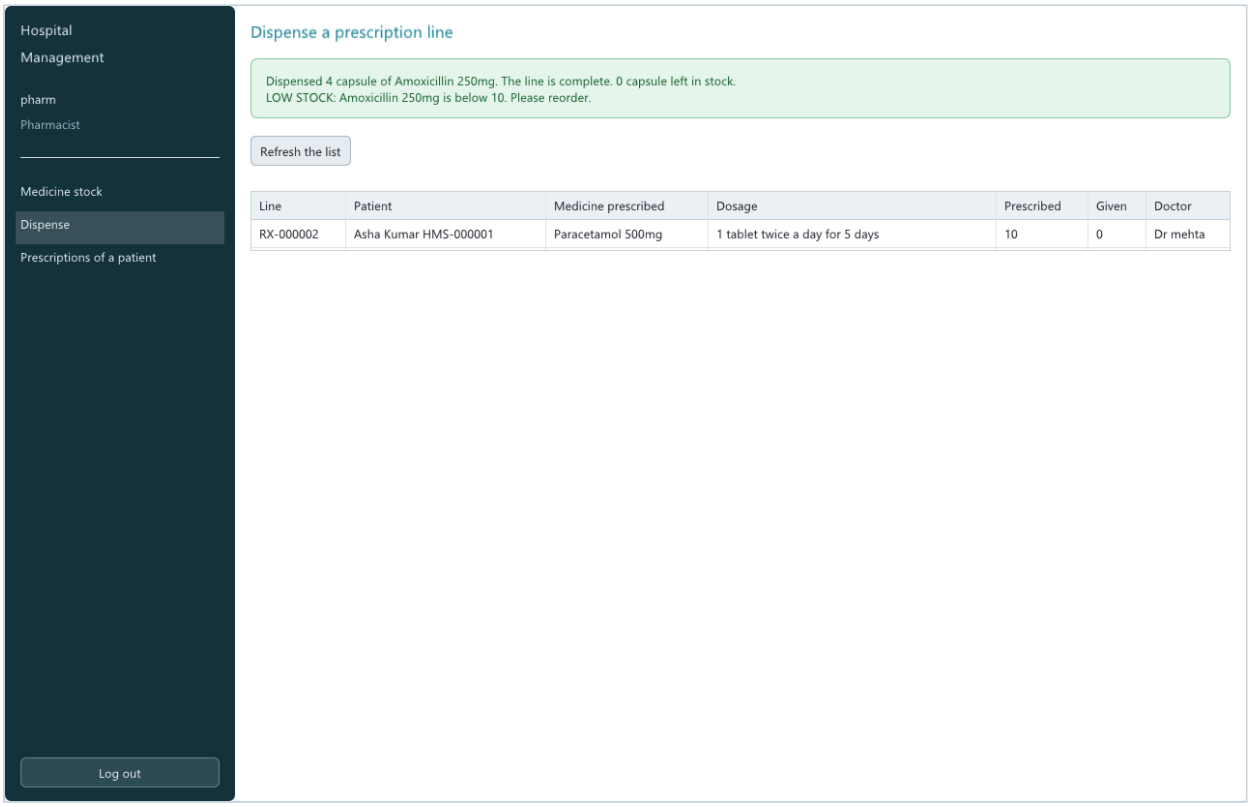
TC-Lab-01 other doctor sees nothing

TC-Pharm-01: Dispensing and stock

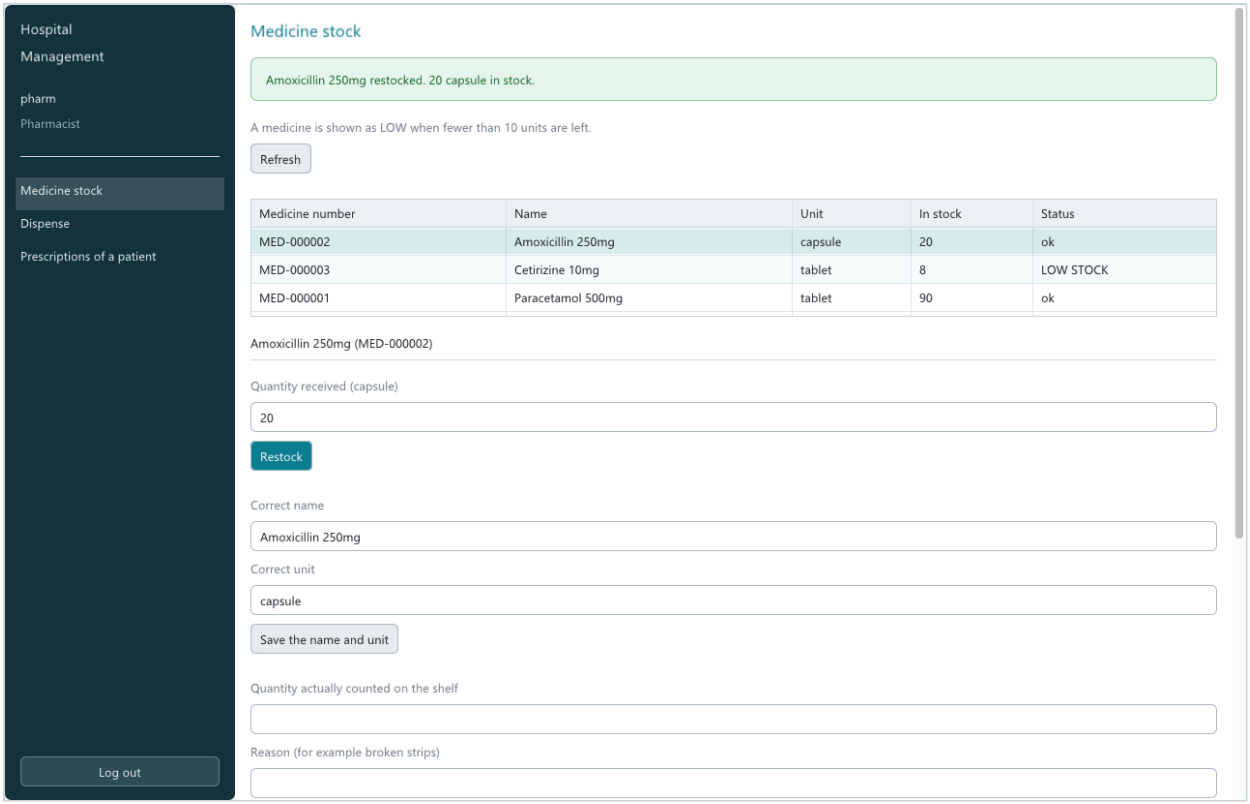
Stock with two LOW STOCK alerts; the warning when the medicine picked is not the one prescribed (the box that allows it is unchecked, and Dispense is disabled); the correct dispense, with a low-stock alert; more than the prescribed quantity refused; a full dispense; and a restock.



TC-Pharm-01 wrong medicine warning



TC-Pharm-01 dispensed with low stock alert



TC-Pharm-01 restocked

TC-Perf-01: Response time

`hms_seed --store perf.db --count 100000 --bench`, 100 runs per operation on a Debug build with an encrypted store. Every operation is far inside the 3 second limit; the tool reports the 90th percentile. The timings cover the service calls, not the drawing of the screen.

```
> hms_seed --store perf.db --count 100000 --bench     (Debug build, encrypted store)

Inserted 100000 patients in 20114.2 ms. Store now holds 100000 patients.
Search by UHID : p50 0.1484 ms, p90 0.2399 ms, p95 0.3367 ms, max 0.704 ms (100 runs; HMS-NF-001 p90 <= 3000 ms: PASS)
Search by name : p50 47.0364 ms, p90 84.6782 ms, p95 86.1843 ms, max 119.15 ms (100 runs; HMS-NF-001 p90 <= 3000 ms: PASS)
Search by phone: p50 0.1515 ms, p90 0.1778 ms, p95 0.2075 ms, max 0.5887 ms (100 runs; HMS-NF-001 p90 <= 3000 ms: PASS)
Booking screen (doctors and free slots): p50 0.9826 ms, p90 1.4506 ms, p95 1.7725 ms, max 2.6571 ms (100 runs; HMS-NF-001 p90 <= 3000 ms: PASS)
Book a slot (find free slot, then book) : p50 7.4962 ms, p90 9.0473 ms, p95 9.6881 ms, max 17.4313 ms (100 runs; HMS-NF-001 p90 <= 3000 ms: PASS)
Note: timings exclude console display and typing.
```

TC-Perf-01 timings

TC-Sec-01: No plaintext PII or medical data

Checked after the full run, on the store the whole run used. The file has no recognisable header and none of the searched strings (patient names, mobile numbers, addresses, clinical text, user names and table names) is readable in it. The audit entries hold IDs and field names only (0 matches for patient text in the audit table).

```
> scan the data store file(s) for readable patient, clinical and account text

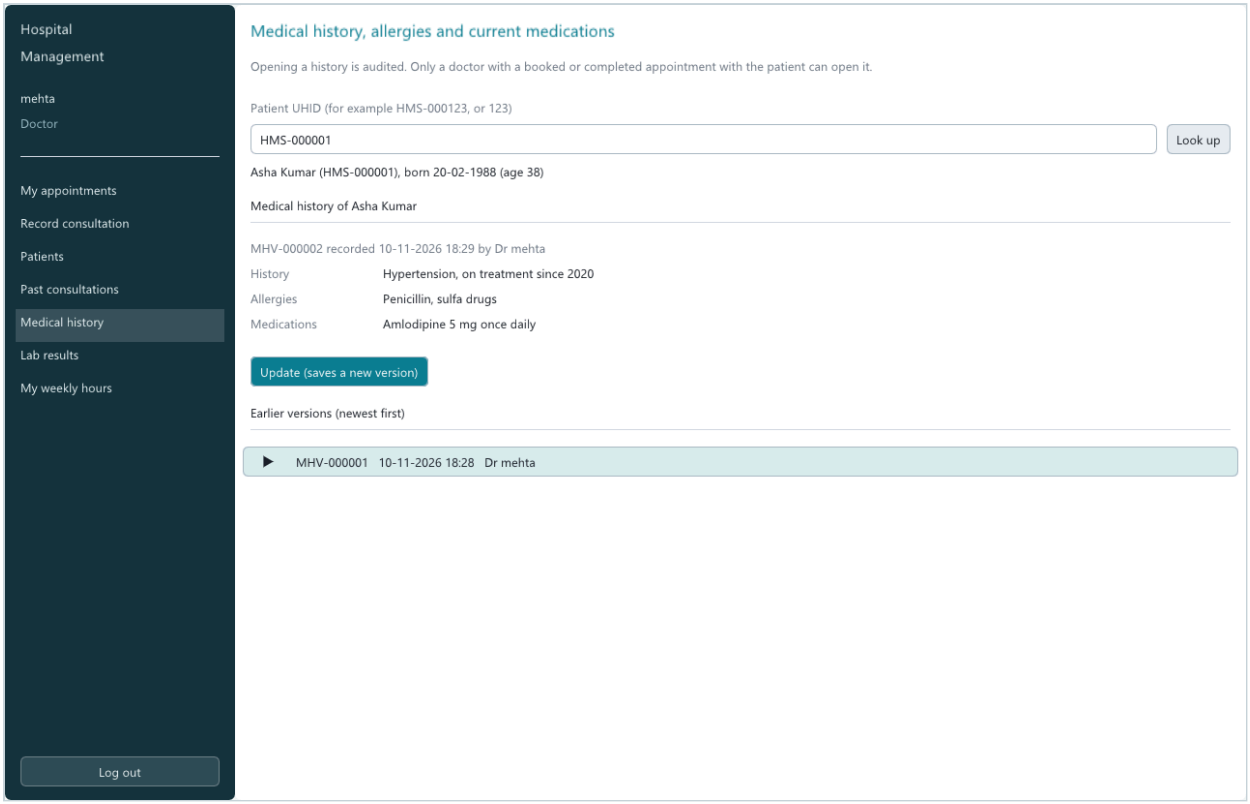
hms.db  282624 bytes
  first 16 bytes: 92 fe 0b 44 dd f2 a3 ce b7 1f 50 df 14 97 39 1d
  Shannon entropy: 7.9994 bits per byte (8.0 would be random)
  readable strings found: none of the 24 searched for
hms.db.lock  0 bytes
  first 16 bytes:
  (empty file)

searched for: Asha Kumar; Meera Shah; Vikram Rao; 91234 56780; 9123456780; 9876543210; 98765 43210; MG Road; Lake View; Palace Road; Bengaluru; Mysuru; Hyperten
sion; Penicillin; Amlodipine; Paracetamol; Amoxicillin; reception; mehta; argon2id; CREATE TABLE; patients; audit_log; SQLite format 3
```

TC-Sec-01 no readable text in the store

TC-Rec-01: Medical history versions

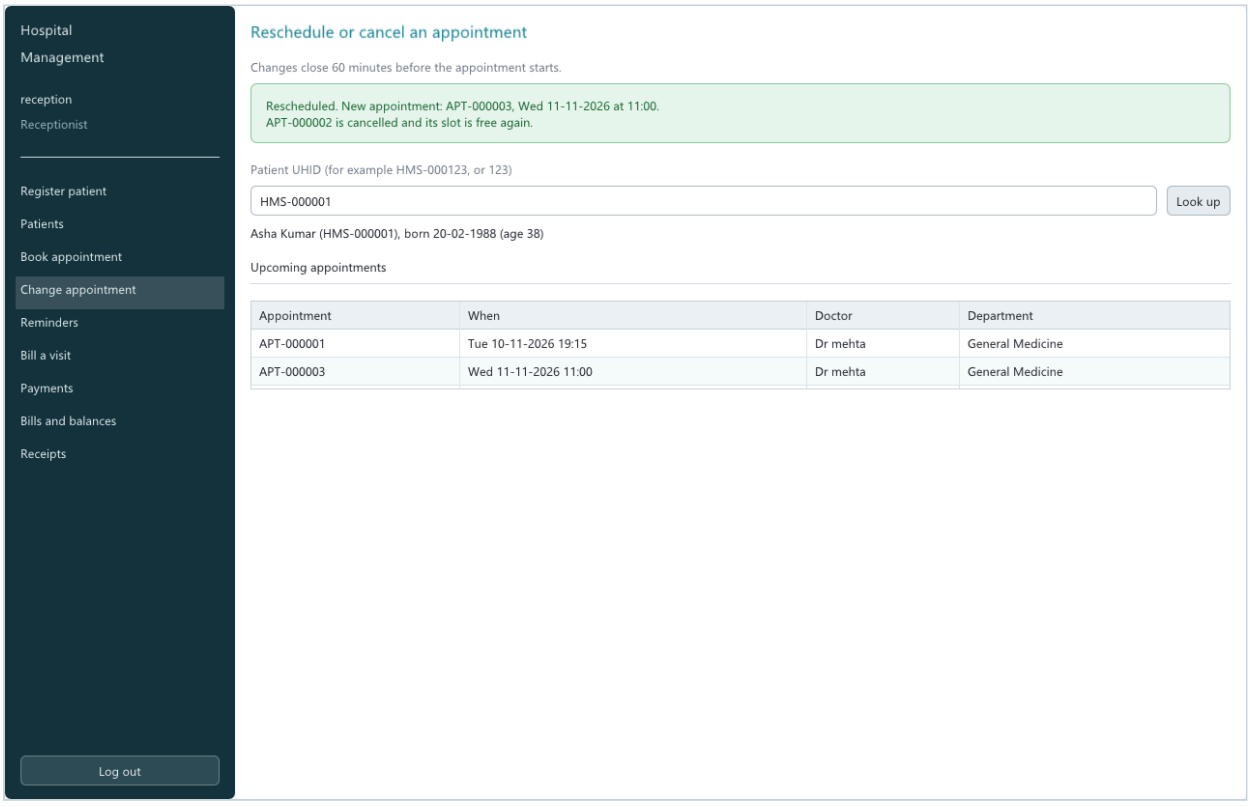
The first version is saved, then only the allergies change. The earlier version stays as it was.



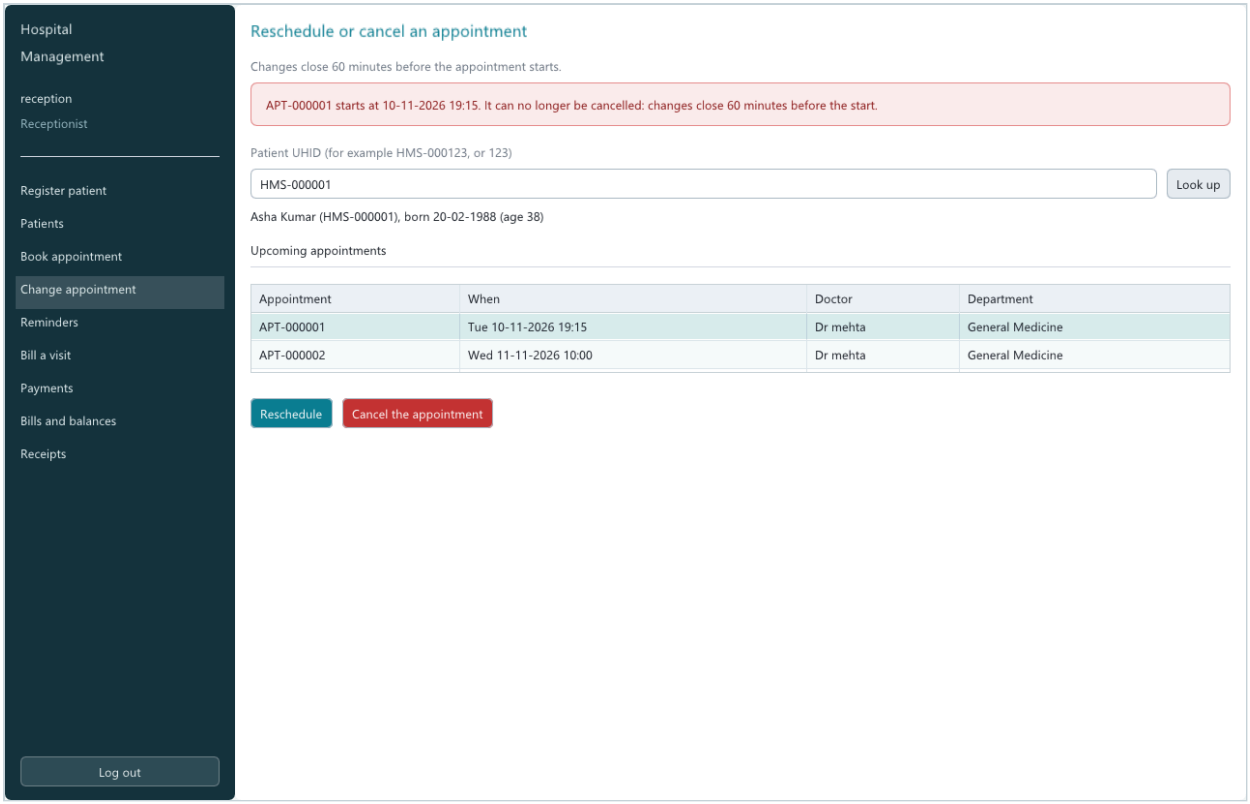
TC-Rec-01 second version

TC-App-03: Reschedule, cancel and cutoff

Reschedule and cancel work outside the cutoff, and the slot is released. A cancel inside the 60-minute cutoff is refused.



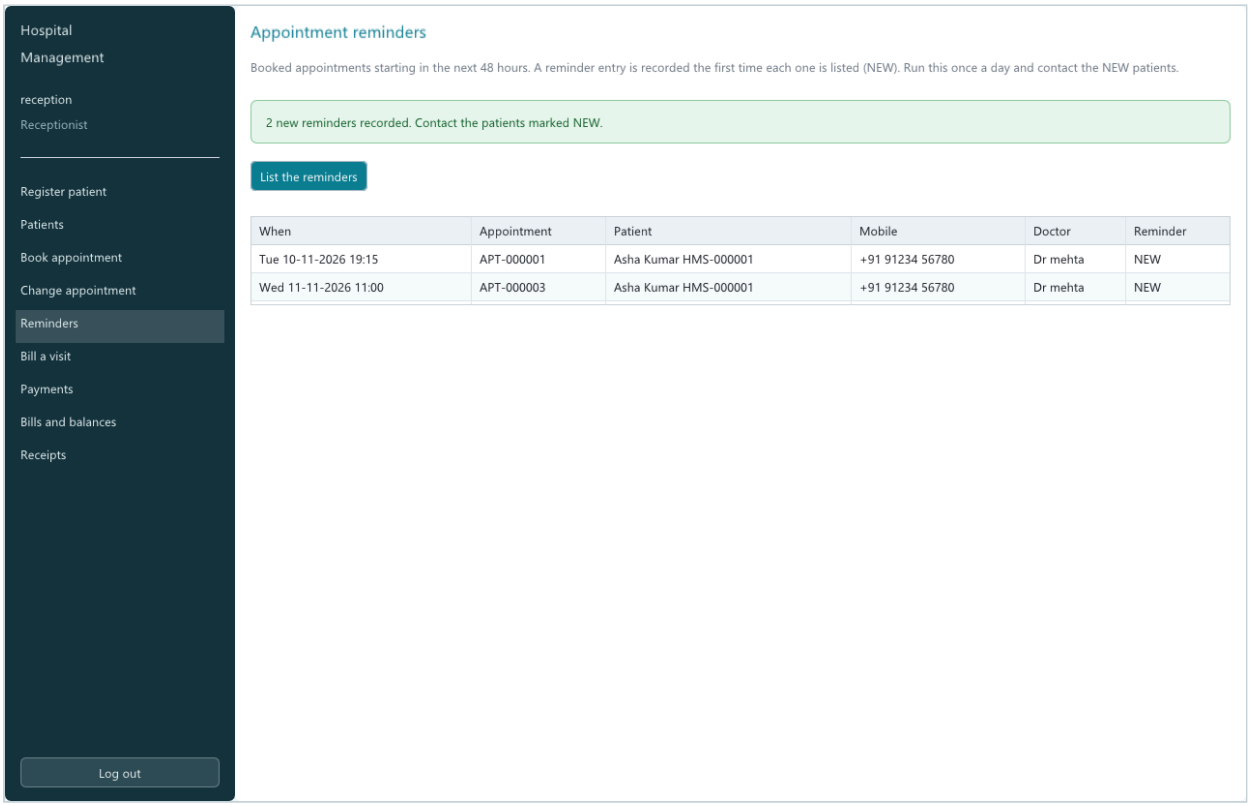
TC-App-03 reschedule



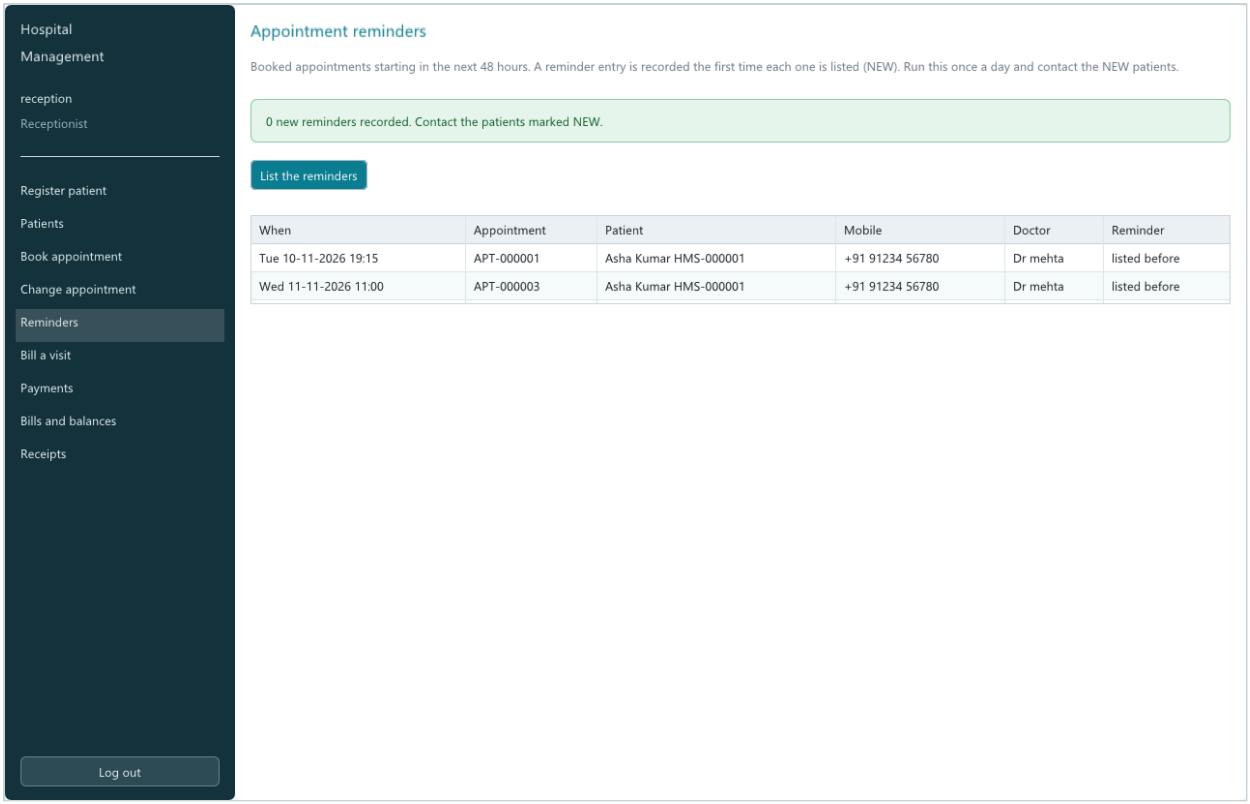
TC-App-03 cancel inside cutoff refused

TC-App-04: Reminders

NEW on the first listing and "listed before" on the second.



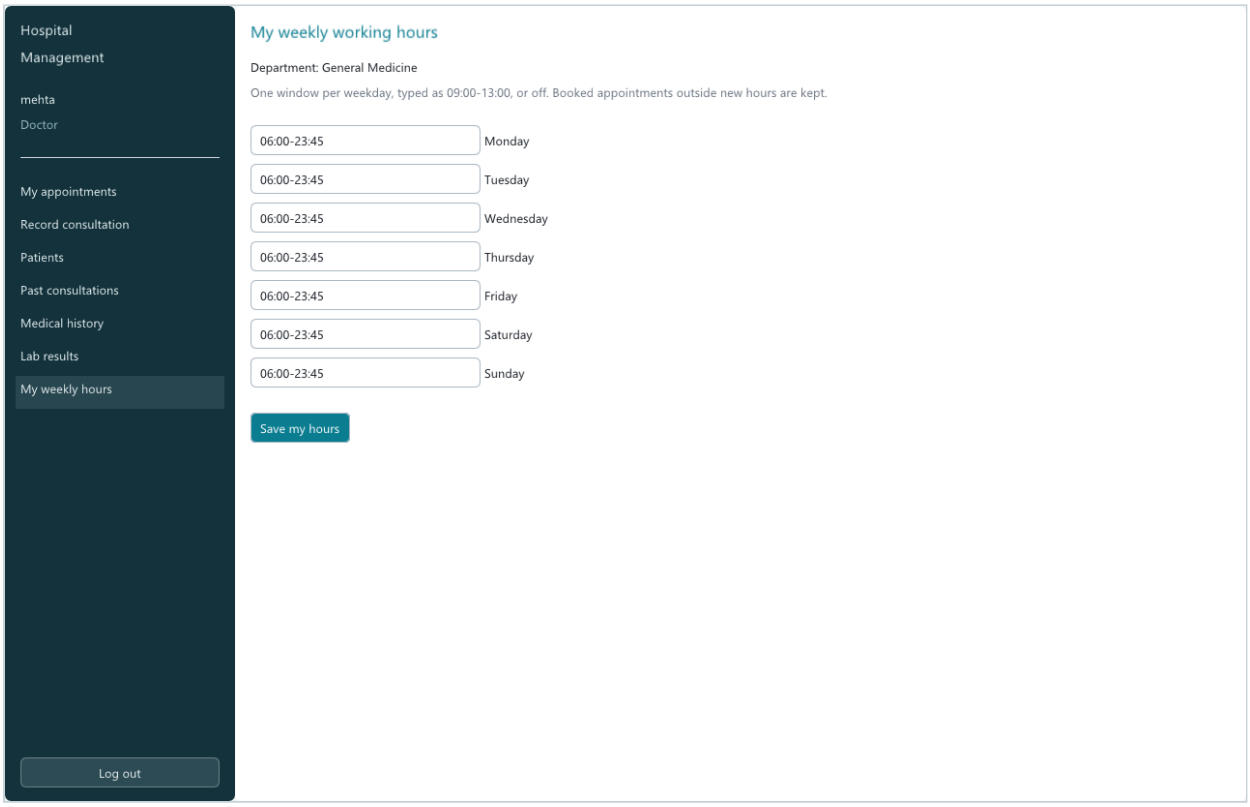
TC-App-04 reminders new



TC-App-04 reminders listed before

TC-Doc-01: Doctor profile

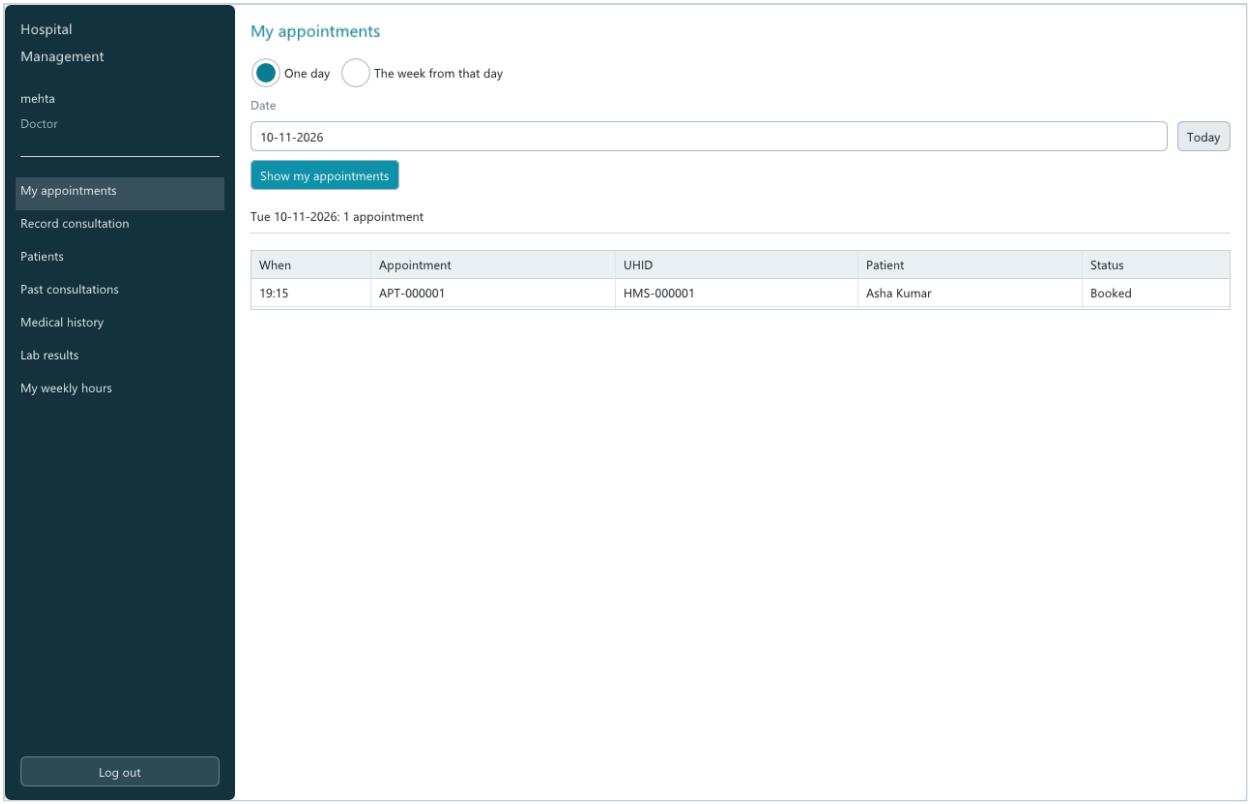
Department and weekly hours of Dr mehta, as set by the Admin. The doctor is offered on the booking screen in TC-App-01.



TC-Doc-01 profile and hours

TC-Doc-02: OPD queue

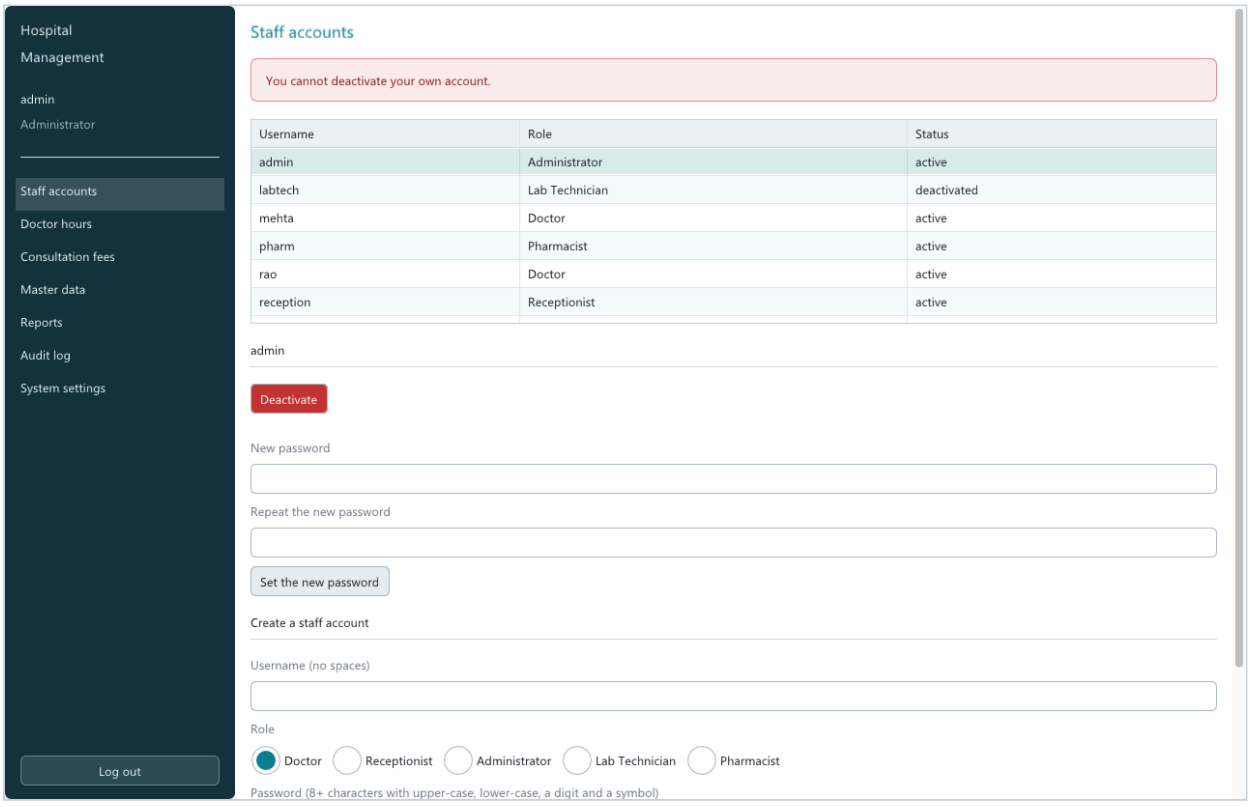
The doctor's own appointments for today.



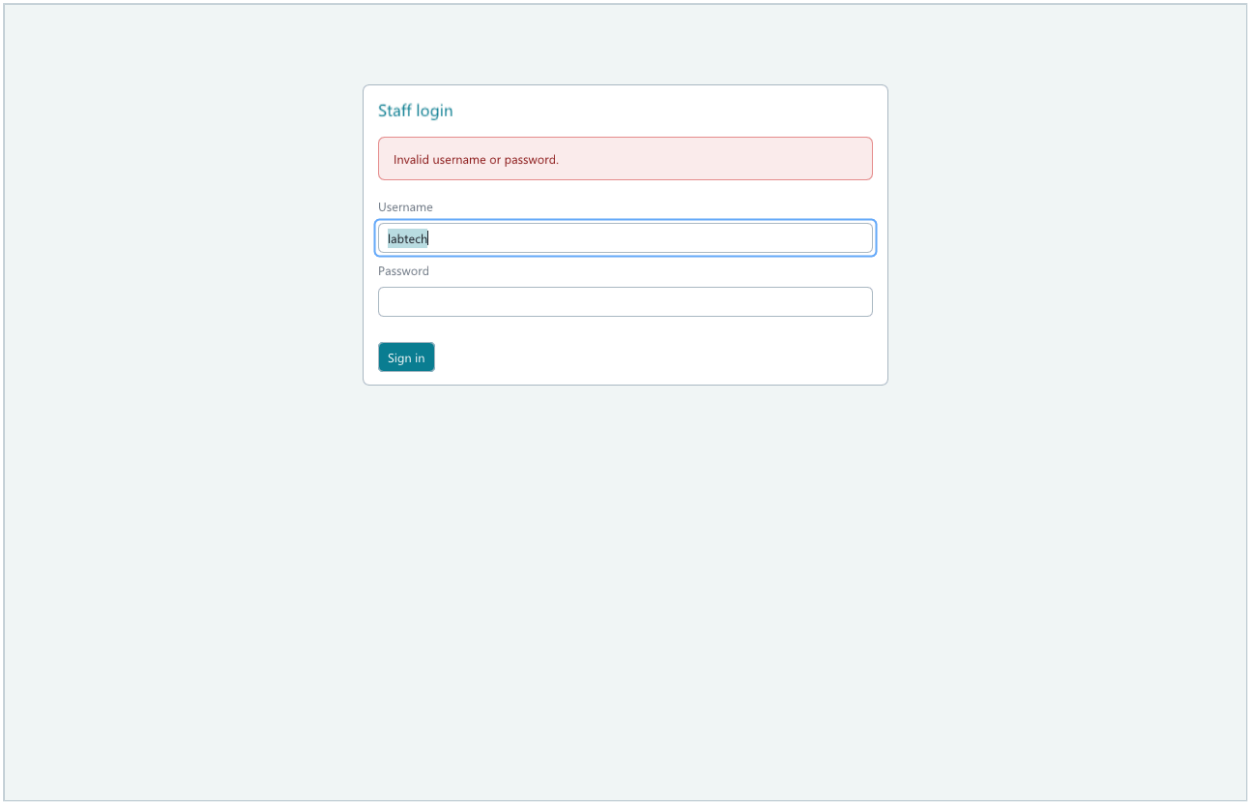
TC-Doc-02 doctor queue

TC-Staff-01: Manage staff accounts

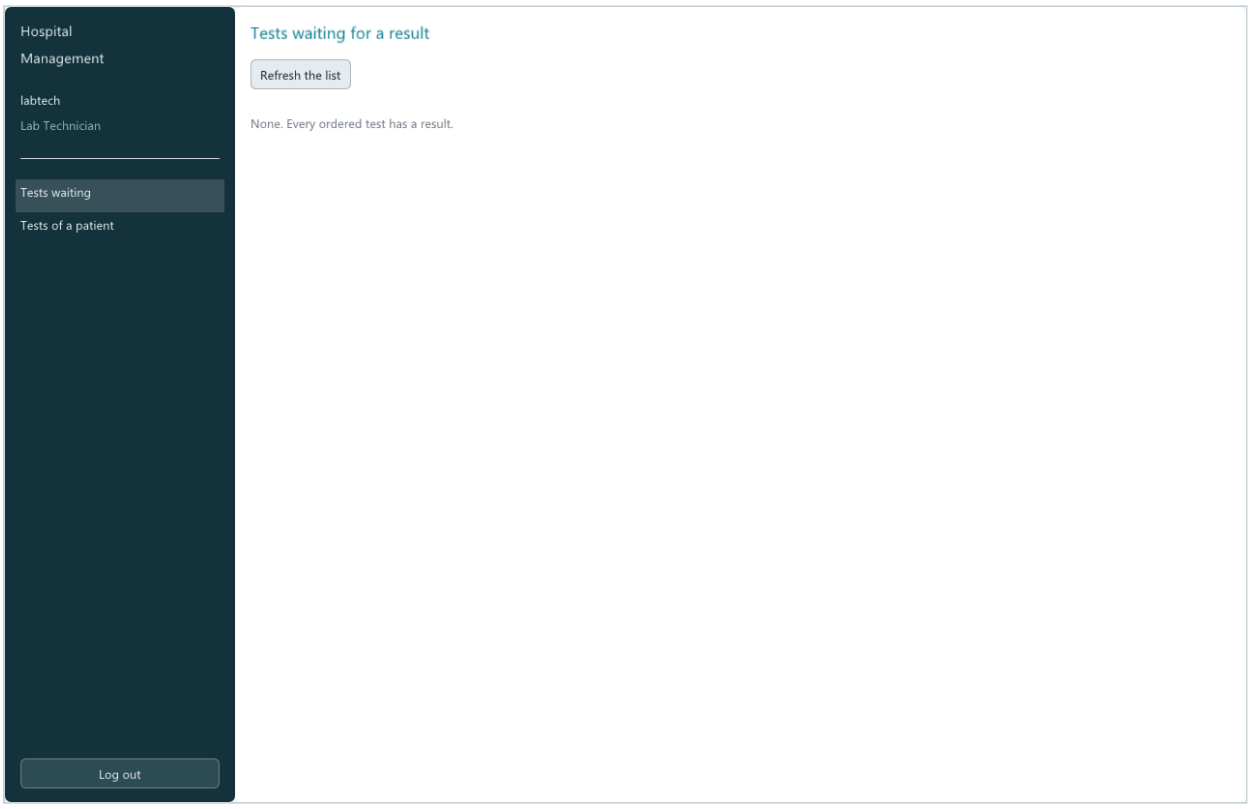
The accounts, a deactivated account, the Admin refused when deactivating their own account, the deactivated account refused at login, then reactivation with a new password and a login with it.



TC-Staff-01 self deactivation refused



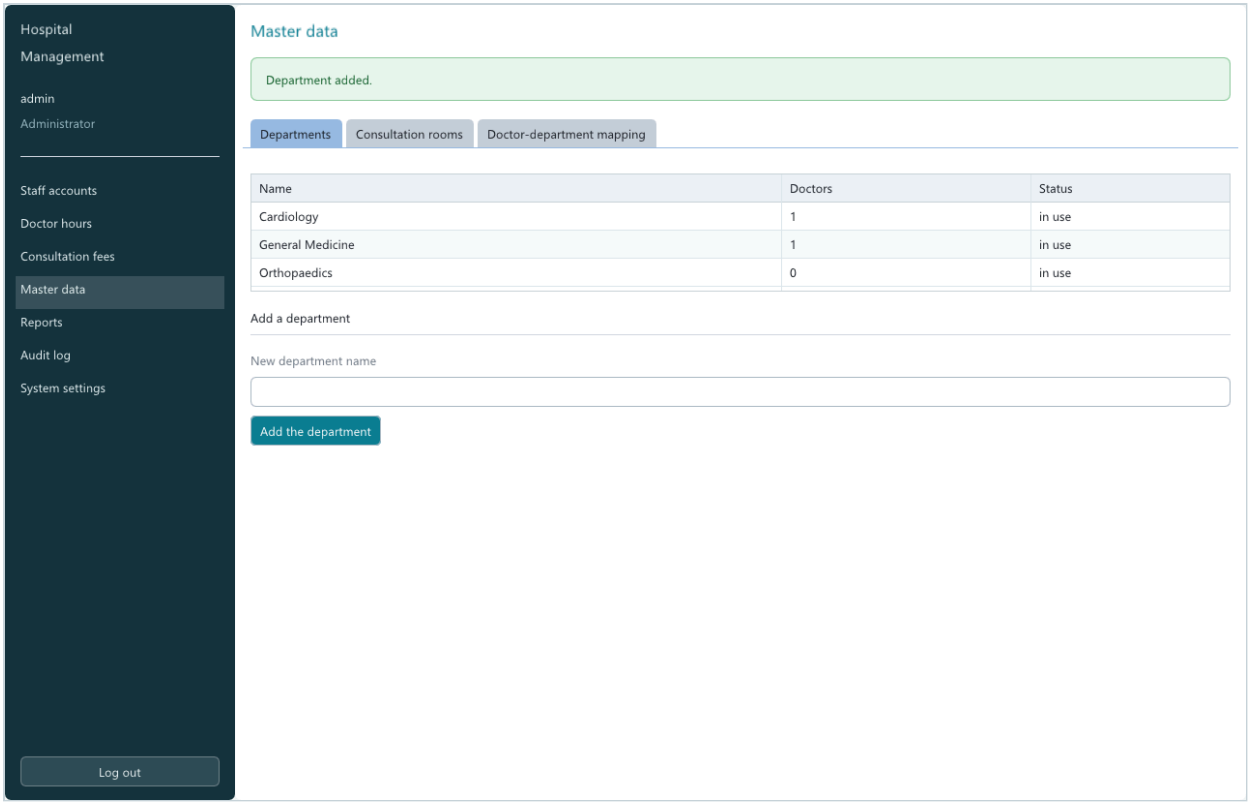
TC-Staff-01 deactivated login refused



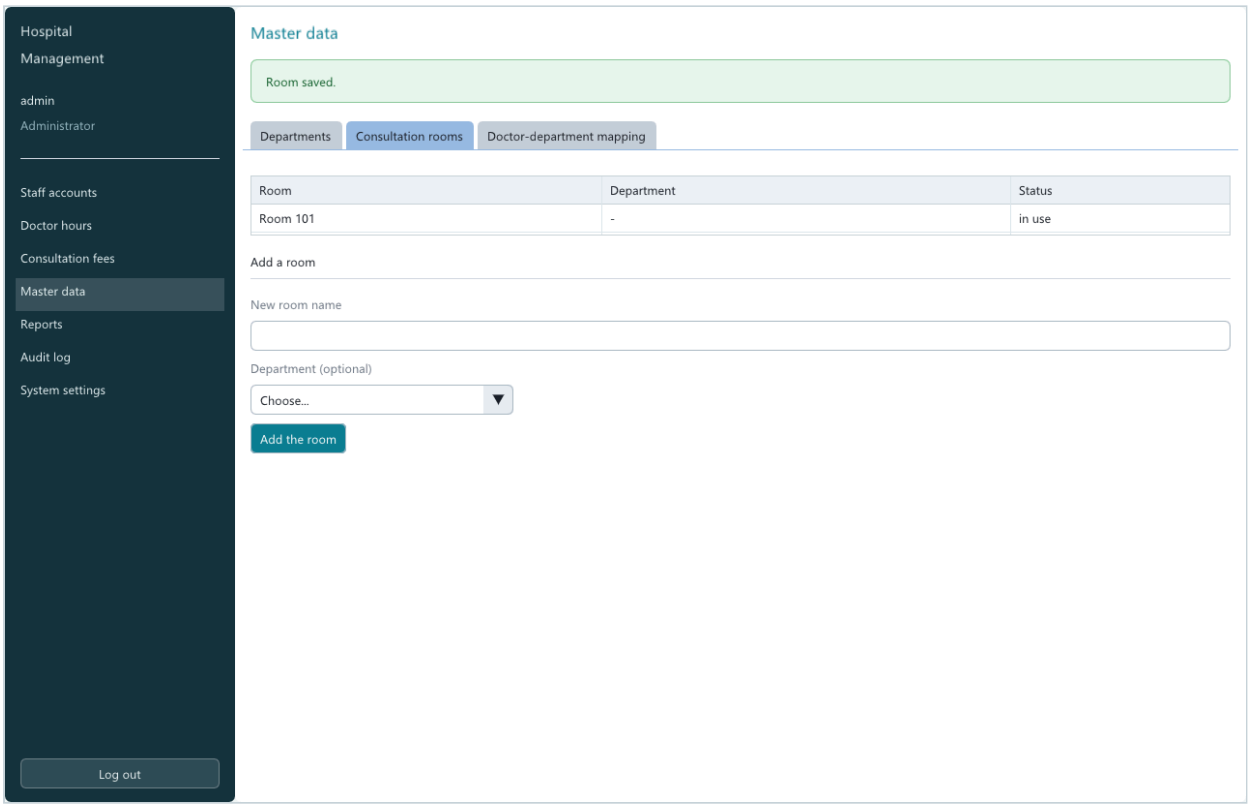
TC-Staff-01 login with the new password

TC-Admin-01: Master data

A department and a room added. The departments of the two doctors appear in the department list, each with one doctor, without the Admin typing them in again (D-59). The mapping lists each doctor's department.



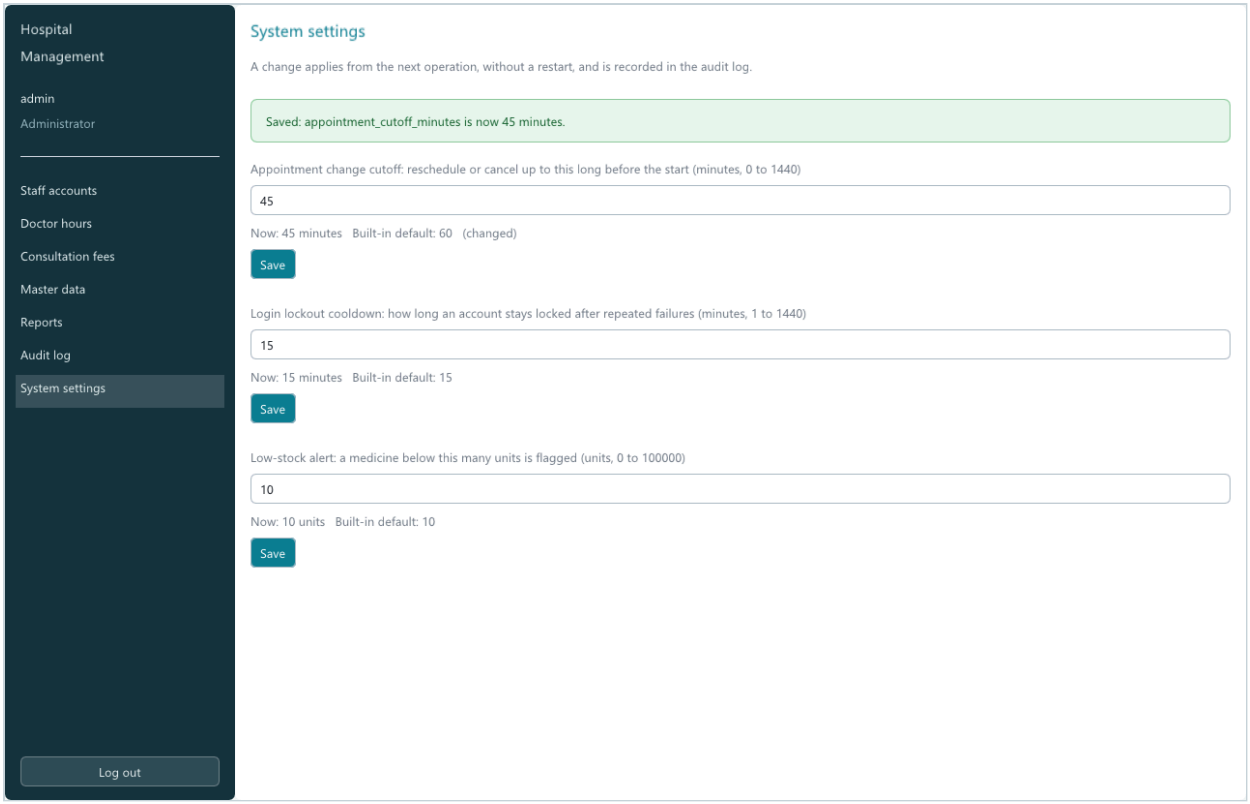
TC-Admin-01 department added



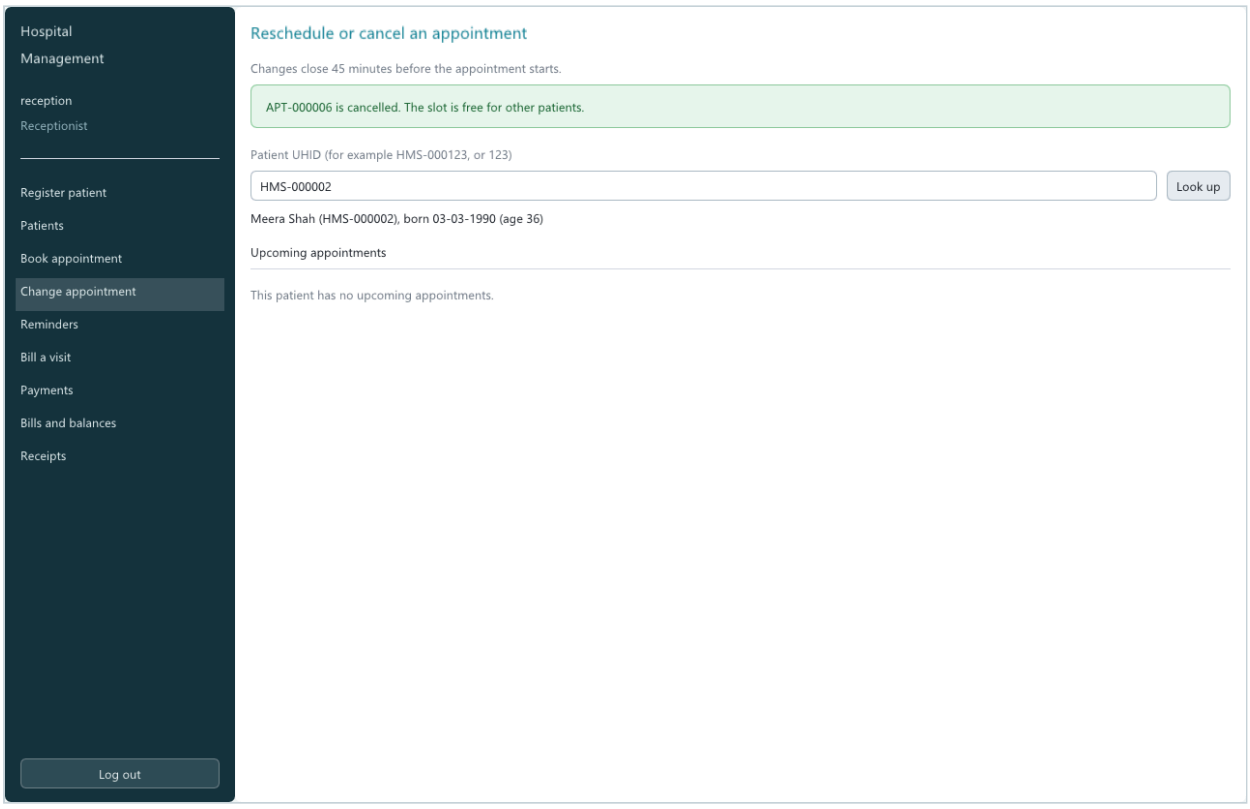
TC-Admin-01 room added

TC-Admin-02: System settings

The cutoff changed from 60 to 45 minutes, and a cancel 59 minutes before the start that the 60-minute rule would have refused, accepted after the change (APT-000006 was booked for 20:30 and cancelled at 19:30:50). The three parameters are listed in the first image.



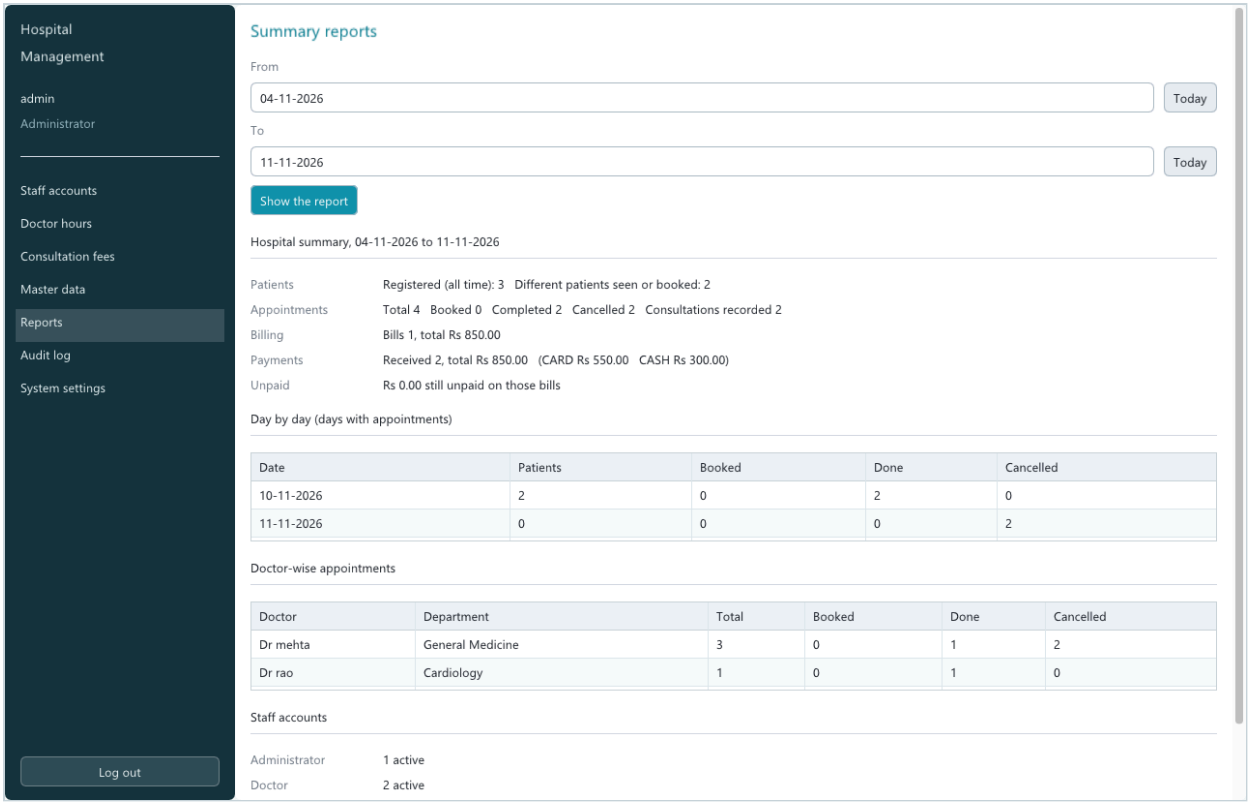
TC-Admin-02 cutoff changed



TC-Admin-02 cancel allowed under the new cutoff

TC-Rep-01: Summary report

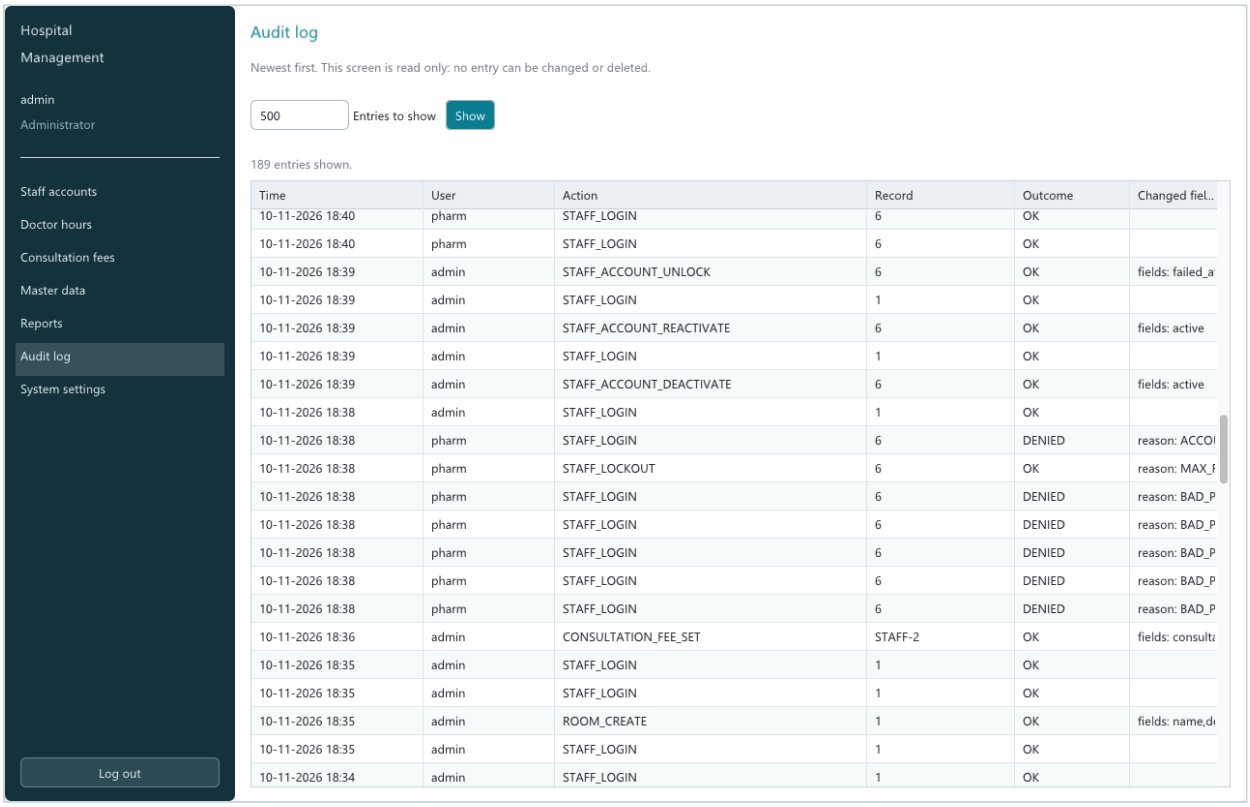
The figures match the data entered in this run (APT-000005 and APT-000006, booked and cancelled for TC-Admin-02 after the report was taken, are not in it).



TC-Rep-01 summary report

TC-Sec-05: Audit trail

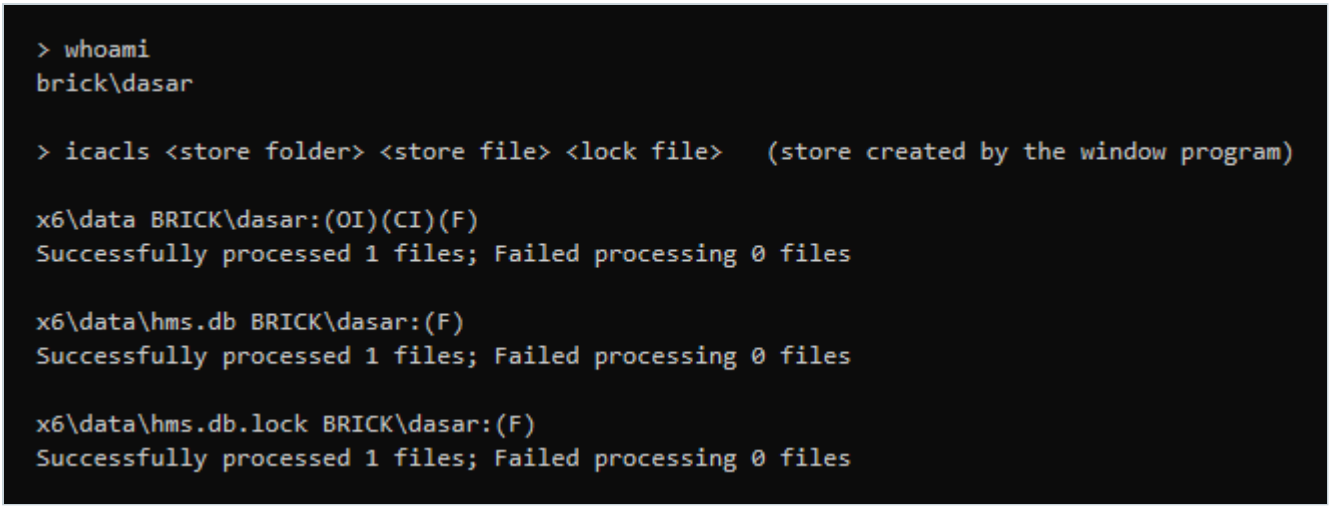
The newest entries, then an older part of the log with the lockout, the denied logins and their reason codes, the unlock, and the account and fee changes. The last two columns are cut off by the window width.



TC-Sec-05 lockout and denials

TC-Sec-06: Data-store access

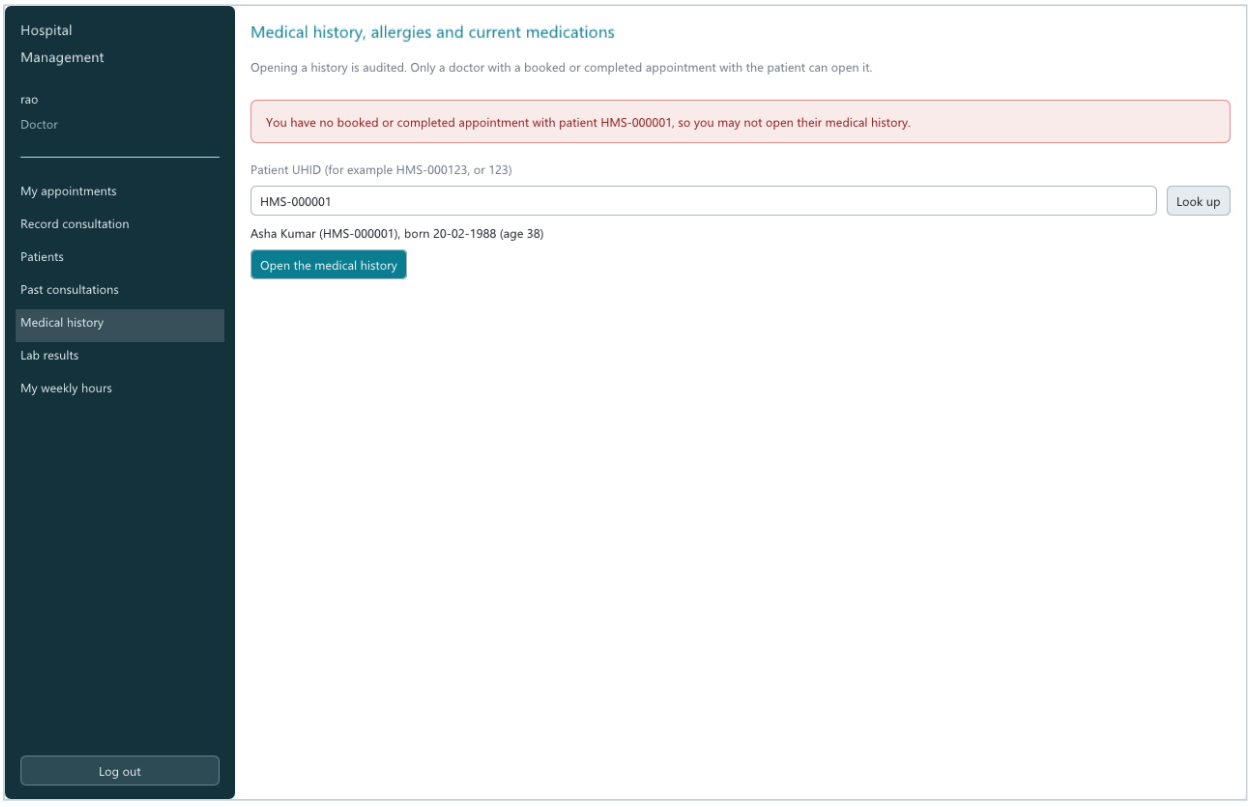
`icacls` on a store created by the window program: the folder, the store file and the lock file each list only the running user. Output of the command, drawn as an image.



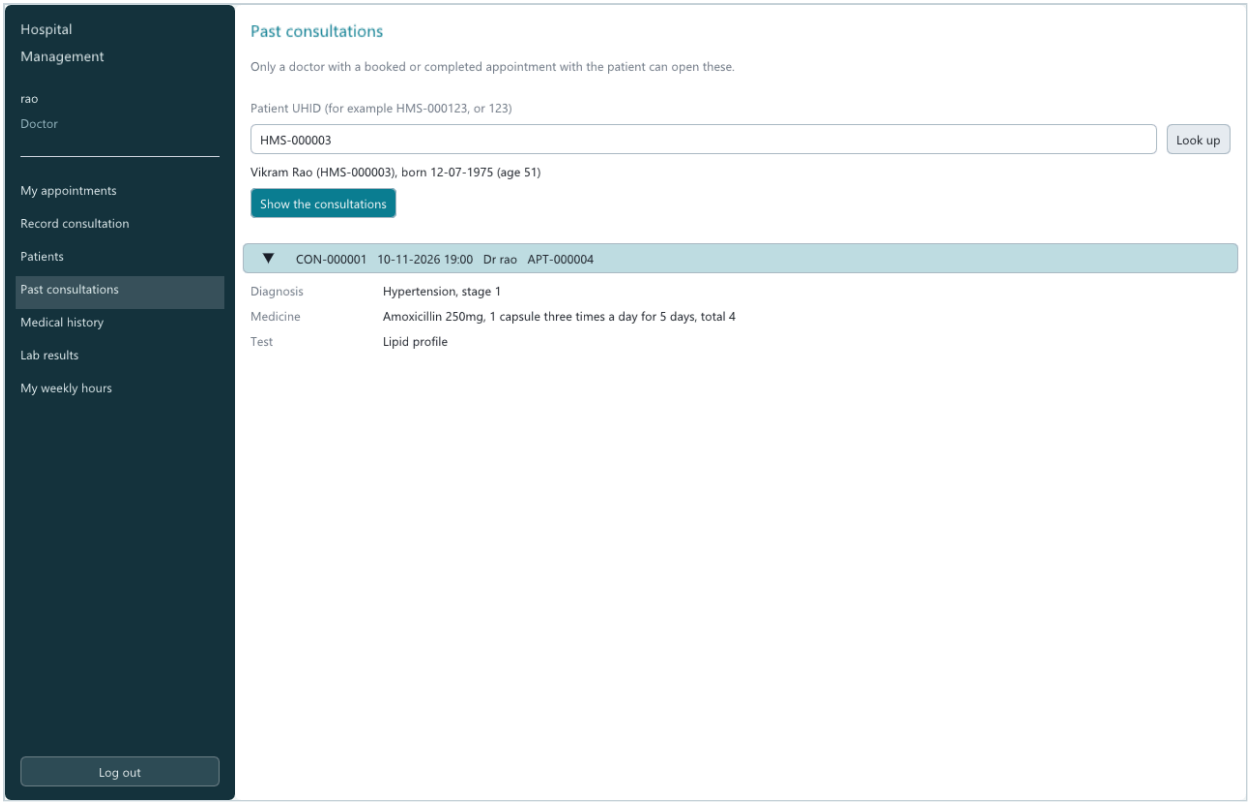
TC-Sec-06 file permissions

X-01: Visit history

A doctor with no appointment is refused both the medical history and the visit history. The treating doctor sees the consultation in full.



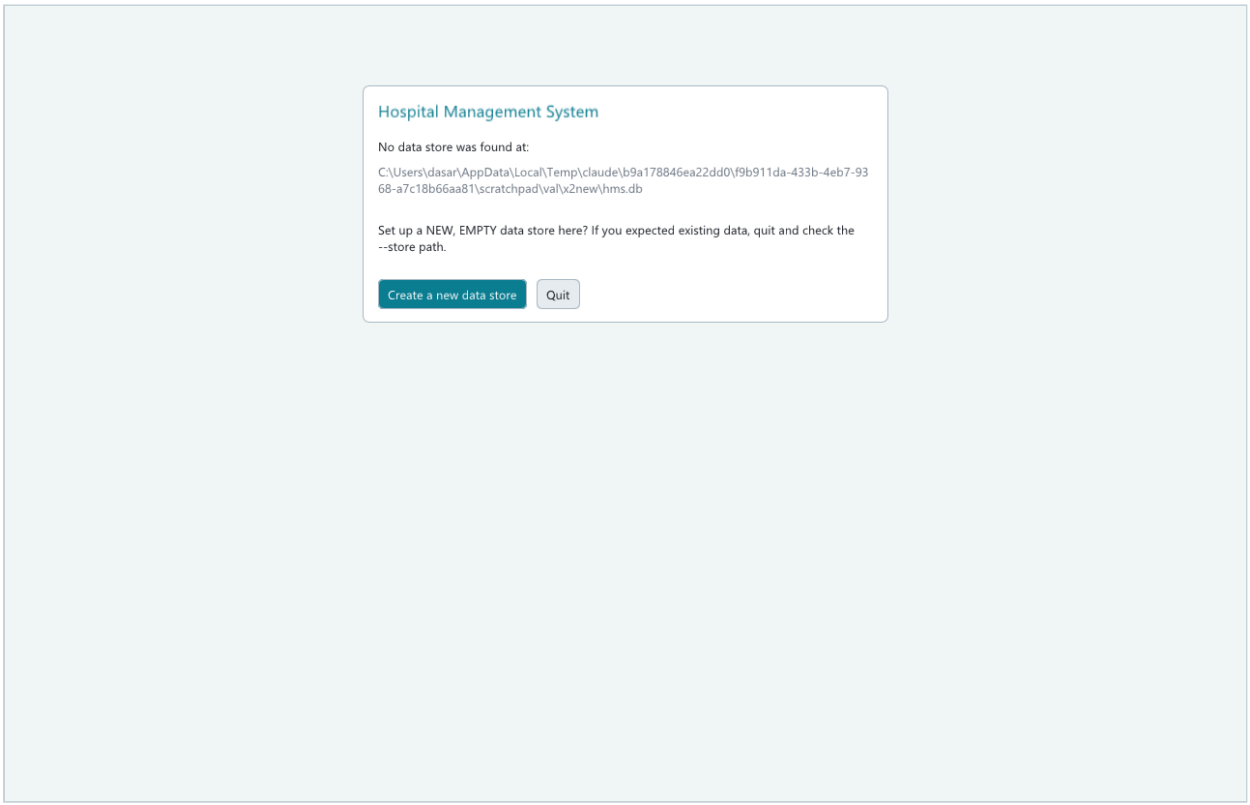
X-01 other doctor medical history refused



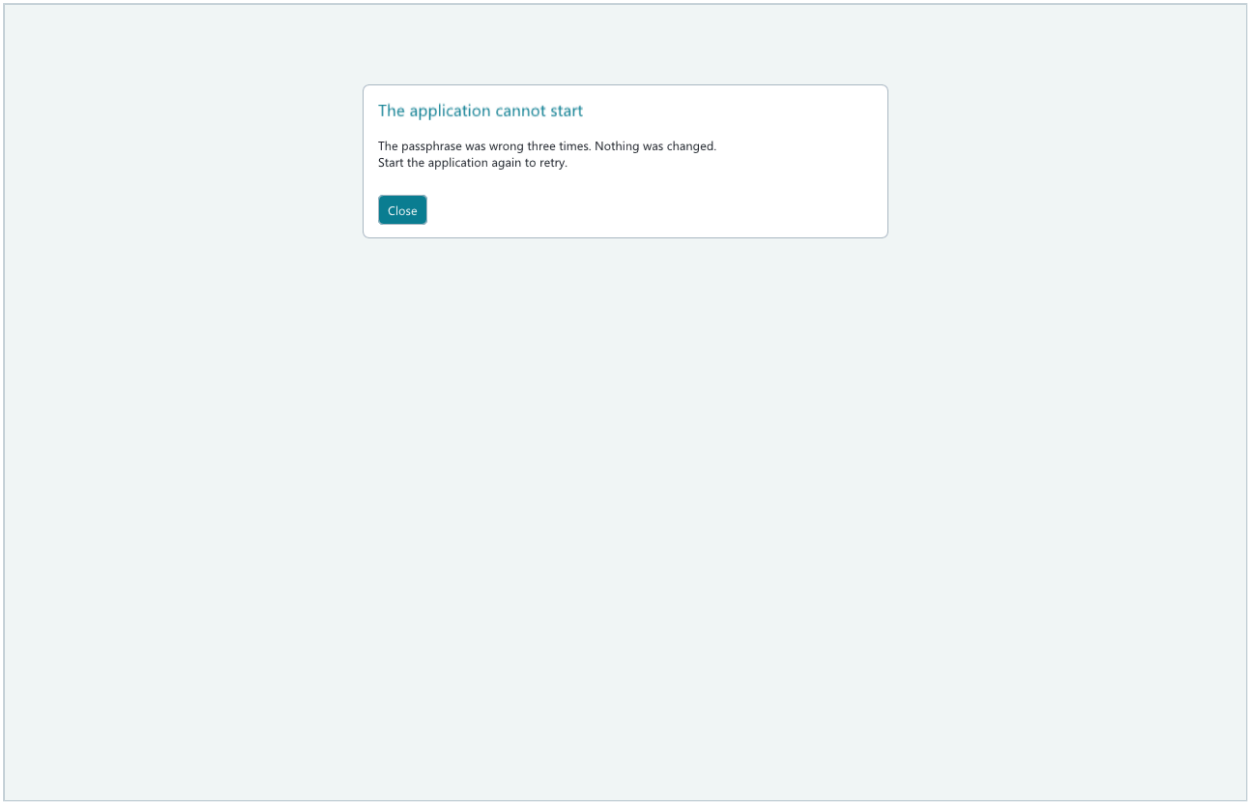
X-01 treating doctor visit history

X-02: Start-up

Declining the new-store question creates nothing. A damaged store is refused after three wrong passphrases and its SHA-256 is the same before and after.



X-02 new store needs confirmation



X-02 damaged store refused

```
> SHA-256 of the damaged data store file, before and after the window program was started on it

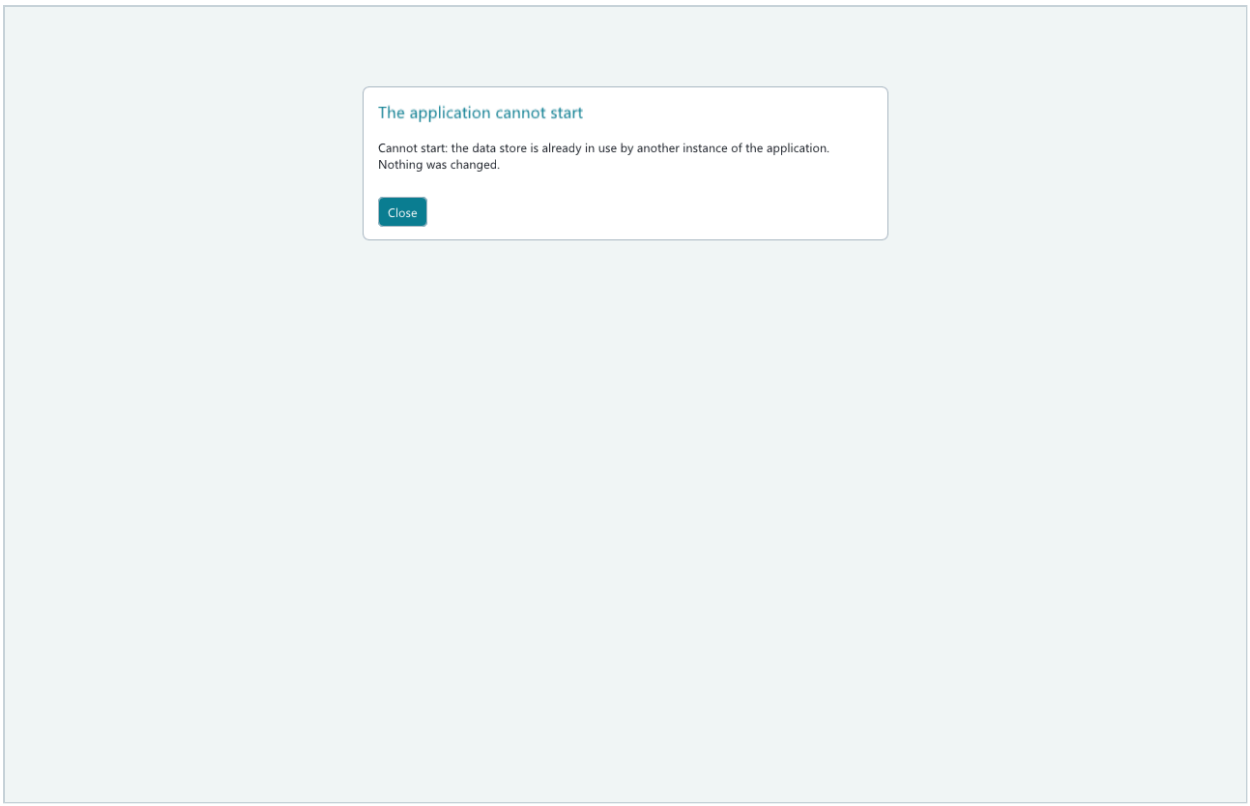
before: 49CB9A57722009BFD287E6F9EADB6BE4BF7F219F8D588A8574708019A313B4AD
after : 49CB9A57722009BFD287E6F9EADB6BE4BF7F219F8D588A8574708019A313B4AD

unchanged: True
```

X-02 damaged store unchanged

X-03: Single instance

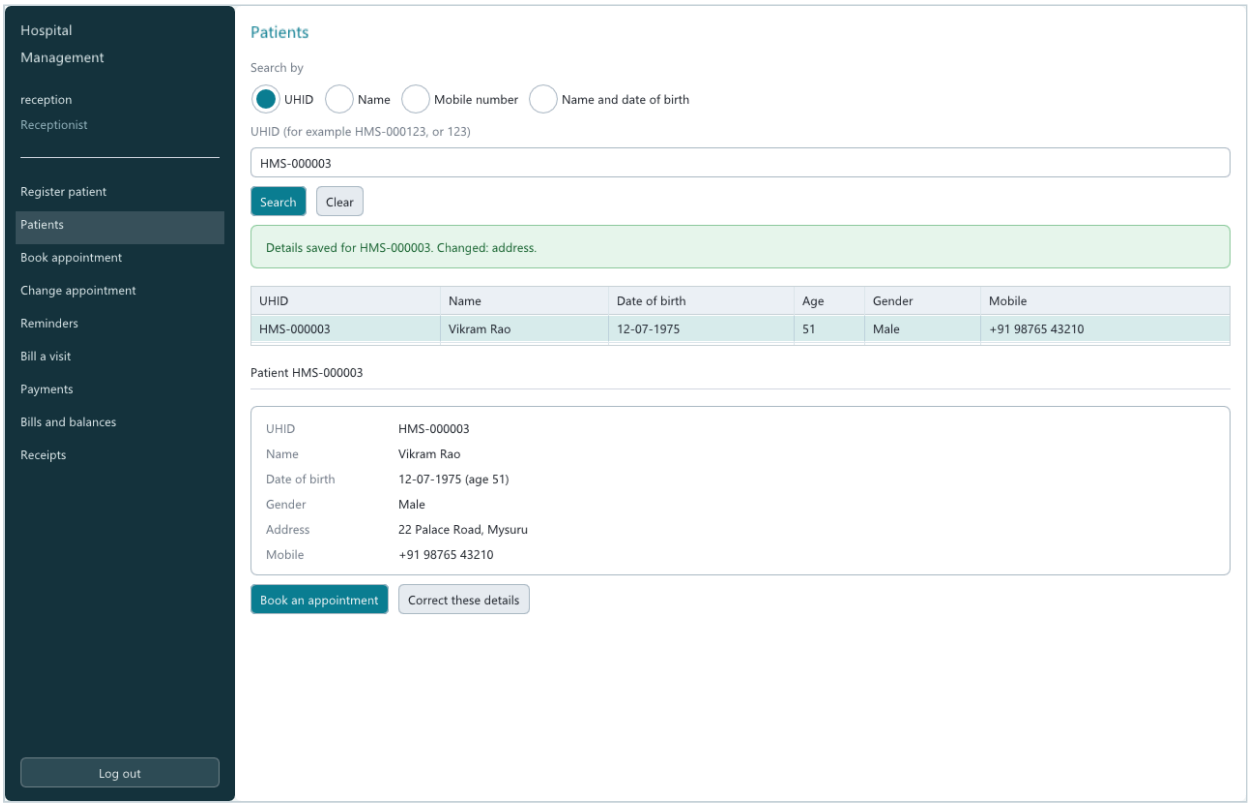
The second instance is refused while the first keeps running. After the first is closed, the next start reaches the passphrase screen.



X-03 second instance refused

X-04: Correct a patient's details

The change is read back before it is saved, then saved. The audit entry names the field (`address`) and not the value.



X-04 saved

Defects found in this run

Both were in the window program, were fixed in the same change, and were checked again afterwards. Neither is in the console program.

- **A refused booking showed no message.** When the chosen slot or the patient's time was already taken, the booking was refused correctly and nothing was saved, but the screen refreshed itself and erased the refusal, so the receptionist saw no explanation. The refusal now stays on screen (the image under TC-App-02 is from after the fix). There is no automated test for the message itself.
- **Lab Technician and Pharmacist could not select a patient.** The "Tests of a patient" and "Prescriptions of a patient" screens looked the patient up as a patient record, which those roles may not view, so the lookup was refused and written to the audit log as a denied `PATIENT_VIEW`. The console menus only check the UHID, as those screens now do. A test now fails if either screen opens the record (`tests/test_gui.cpp`).

Observations that are not defects

- The window program shows a receipt on screen and cannot save it as a text file, which the console program can (D-46, D-61).
- Text in scripts the font does not contain, for example Devanagari, would show as "?" (D-61).
- The last two columns of the audit log table are cut off at the default window width.
- The reminder screen says "Contact the patients marked NEW" also when 0 new reminders were recorded.
- Timings were taken on a Debug build and exclude drawing the screen.

Known limitations

Known limitations and agreed differences from the SRS, SAD and Test Plan are recorded in `docs/deviation-log.md`.